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Original article

Site location and allocation decision for onshore wind farms, using spatial multi-criteria analysis and density-based clustering. A techno-economic-environmental assessment, Ghana

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ABSTRACT

This paper presents a new approach for identifying and site contour optimization of wind farms in the context of transmission expansion planning for the Republic of Ghana to support its renewable energy development plan. The proposed approach uses spatial multi-criteria analysis, density-based clustering, and analytical hierarchy process to identify, optimize, and rank candidate sites. The proposed methodology provides an automated procedure for optimizing site boundary contours using density-based clustering. It provides decisional flexibility in identifying clusters of the minimum required size, unlike the traditional approach. The analysis identifies 14 geographical clusters with high wind energy availability with an average area of 19 km^2 and a maximum area of up to 32 km^2. All the clusters found are in relative proximity to both transportation and transmission networks. Results from the techno-economic and environmental assessment identified the least levelized cost of energy of 0.21 $/kW h at clusters C, E, M and N. The power plant modeled at cluster M recorded the least simple payback period of 4.30 years and highest internal rate of return of 22.8 %. The worst site was cluster L because it recorded the highest emissions of about 354,474 kg/year for carbon dioxide and 130 kg/year for carbon monoxide.

Introduction

Energy availability is a strategic factor for every country relative to its economic growth and social development [1–4]. Fossil fuel (i.e., oil, coal, and natural gas) constitutes more than 80% of the world's energy generation [5,6]. However, the debilitating effects caused by fossil fuel-based energy generation on the environment coupled with climate change have pushed governments to look for more clean energy. This has led to an increase in interest in various renewable energy (RE) power systems in the last 30 years [7–9]. Developing countries, compared with developed countries, face different development dilemmas. They need to plan for their infrastructure expansion and provide sustainable and affordable energy to comply with sustainable development goals.

Wind energy is currently among the world's top exploitable RE resources, along with solar and tidal energy. It leads the global market principally because of its high availability, technological maturity, cost competitiveness, and macro-level data availability [10]. The total cumulative wind power capacity globally increased from 24 GW in 2001 to over 591 GW in 2018 and is projected to exceed 908 GW by 2023. China reached a cumulative installed capacity of 211 GW in 2018, making it the leader in wind energy generation globally, followed closely by the United States with 96 GW and India with 35 GW [11].

One of the significant limitations in introducing RE in developing countries has been the lack of reliable data about various available renewable resources. The international development agencies have partly addressed this limitation by compiling the macro-data necessary for resource evaluation and planning. However, comprehensive resource availability analysis and the optimum location of the resources, and additional infrastructure development assessment are required to tap these resources. The limited availability of such assessments has been why developing countries are slow in developing RE systems on a large commercial scale. Investors rely on such assessments to evaluate the techno-economic potentials of locations during planning [2].

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Nomenclature

AHP	Analytical Hierarchy Process
CF	Capacity Factor
CI	Consistency index
COPRAS	Complex Proportional Assessment
CR	Consistency Ratio
CW	Criteria Weight
DBC	Density-Based Clustering
DBCA	Density-Based Clustering Approach
DG	Diesel Generator
FANP	Fuzzy Analytic Hierarchy Process
FTOPSIS	Fuzzy Technique for Order of Preference by Similarity to Ideal Solution
RE	Renewable energy
GIS	Geographical Information System
GW	Gigawatt
MCDM	Multi-Criteria Decision Making
SMCA	Spatial Multi-Criteria Analysis
TRI	Terrain Ruggedness Index
WGS	World Geodetic System
WPP	Wind Power Plant

Several studies have used different approaches to identify suitable sites to construct wind or solar power plants using different methodologies. Some of the approaches include but are not limited to multi-criteria decision-making (MCDM). The MCDM is an authoritative decision-making tool that can help deal with problems in selecting appropriate wind farms. It helps eliminate uncertainties and biases that may be associated with a decision made. It is a suitable tool for planners who encounter challenges during selecting appropriate sites from several alternatives based on the site attributes [12]. Studies on wind energy site selection using GIS and MCDM are in [13–23]. Most of these studies identified sites in the proximity of the transmission network, which is a good approach if the country has an adequate transmission network system, which is rarely the case in developing economies. As a result, developing countries need to incorporate future transmission network expansion in their decision-making for developing new power generation projects.

In other studies, Liang et al. [24] proposed a probability-driven robust transmission expansion planning (TEP) to help obtain the best investment strategy relative to RE in China. Jadidoleslam et al. [25] developed a framework for expanding transmission and wind power planning which modeled a stochastic bi-level optimization problem. Veronesi et al. [26] used statistical analysis to quantitatively evaluate weights for GIS analysis for transmission lines siting decisions in Switzerland. Fernandez-Jimenez et al. [27] proposed a new approach to rank the candidate sites to develop photovoltaic power plants according to their visibility. They employed distance decay and fuzzy viewshed methodology in their analysis.

Doljak and Stanojević [28] integrated the GIS and MCDM methodology to select the optimum site for developing PV power plants. [29], also used the MCDM and GIS approach to choosing the right location for constructing a wind farm. Furthermore, Villacreses et al. [30] employed the MCDM-GIS methodology to select the most feasible location for installing a wind power plant in Ecuador. The optimum site for developing a wind-powered hydrogen refueling station in Adrar, Algeria, was assessed using the AHP-GIS approach [31]. Sánchez-Lozano et al. [32], combined the Fuzzy Analytic Hierarchy Process (FAHP) and the Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (FTOPSIS) with the GIS to ascertain the optimum site for onshore wind farms. Konstantinos et al. [33] employed the AHP and TOPSIS for wind energy site selection in Greece. A regional-scale wind farm site assessment in Greece was performed by [34], using the GIS-MCDM approach. Dhiman and Deb [35] also integrated Fuzzy TOPSIS and fuzzy Complex proportional assessment (COPRAS) to assess hybrid wind farms.

Also, Saraswat et al. [36] developed a GIS and MCDM technique to evaluate wind and solar farms in India. Their study identified Rajasthan state in India as the region with the highest suitable land for the development of both wind and solar farms. Giamalaki and Tsoutsos [37] presented a methodology that combines AHP and GIS to clarify and prioritize suitable locations for solar farms in the Mediterranean. According to their results, a maximum of 530 MW and 30 MW of photovoltaic and concentrated solar power, respectively, can be obtained from the highest priority areas. Brewer et al. [38] also assessed appropriate site for the installation of solar project in southwestern United States using the GIS-based MCDM approach with an exclusive social preference component. Their study found out that the inclusion of the social preference component to the model had a significant impact on the amount of suitable areas, it led to a reduction in the areas identified. Vafaeipour et al. [39] evaluated areas in Iran that are suitable for the implementation of solar farms using a hybrid MCDM approach. Finally, Wang et al. [40] mapped the suitable locations for the development PV power plants in Tibet, China. They implemented the GIS overlay taking into account the local terrain, solar energy distribution and the land cover of the area.

A critical analysis of the applied methodologies in the reviewed literature supra highlights biases at the initial stages. These methods often favor infrastructure proximity resource development rather than basing the decision on the generation capacities of the sites, which skews the judgment towards less development cost at the detriment of overlooking the optimum decision.

The existing methodologies are primarily suitable for countries with mature and efficient energy transmission networks and favor certain parts of countries already close to the significant infrastructure, which is a substantial disincentive to rural development. However, the transmission networks are underdeveloped in most developing countries, and they suffer from energy transmission inefficiencies. As a result, these countries must develop new energy generation resources and simultaneously plan for future transmission network expansion. Such planning requires a complete road map of the top exploitable sites, and our proposed methodology is helpful in this context.

Due to the recurring energy crisis confronting the Ghanaian people, the government of Ghana has planned to increase the country's renewable energy generation to 10% by the close of 2030 [2], including wind and solar energy generation due to the enormous potential of these resources in the country. Therefore, this study plans to identify wind hotspots in Ghana using a novel hybrid methodology for site selection of wind power plants using spatial multi-criteria analysis and density-based clustering. This work extends Ref [6] but uses wind data and incorporates the multiple criteria relevant to the wind farm site selection. This work also demonstrates the usability of classical tools for techno-economic and environmental assessment.

The proposed site selection methodology uses a GIS-based approach to eliminate the unsuitable locations, followed by applying the density-based clustering (DBC) on the remaining candidate sites to pinpoint the top energy available areas to develop wind power plants. The proposed methodology is demonstrated in Ghana and validated by comparing our findings to the reports of international development agencies. For the first time, this paper incorporates spatial multi-criteria with density-based clustering for final site selection decisions of wind power projects.

The critical contribution of this work includes the provision of a new methodology for identifying the top exploitable RE resources at the national level to facilitate the development of RE resources and the concurrent planning of transmission network expansion. The complete techno-economic-environmental study of the Wind/DG/Battery/Converter system for all specified clusters confirms the proposed methodology's compatibility with the available tools for the project's

Table 1
Selection criteria for wind farms location [44].

Factor	Suitability Ranking			
	Highly Suitable	Moderately Suitable	Low suitability	Not suitable
	3	2	1	0
Wind speed (m/s)	>6	>5 & ≤6	>4 & ≤5	<4
Proximity to roads (km)	>0.5 & ≤2	>2 & ≤5	>5 & ≤10	>10
Proximity to transmission lines (km)	>0.5 & ≤2	>2 & ≤5	>5 & ≤10	>10
Land use	Barren grassland	Agricultural land	Short vegetation and shrubs	Public settlements, wetlands, airports.
Slope (%)	≥0 & ≤7	>7 & ≤12	>12 & ≤15	>15
Distance to airports (km)	>4	>3.5 & ≤4	>3 & ≤3.5	≤3
Distance from residential area (km)	>3	>2 & ≤3	>1 & ≤2	≤ 1

feasibility. HOMER Pro software is used for this analysis Ref [41–43].

The paper is organized into four sections. Section “Methodology” presents the study methodologies, i.e., spatial multi-criteria analysis, density-based clustering, AHP for site ranking and sensitivity analysis, techno-economic and environmental analysis. Section “Results and Discussions” presents the results and discussions. The conclusions and policy implications are discussed in Section “Conclusions and policy implications”.

Methodology

This paper proposes a hybrid methodology for selecting wind power plants using spatial multi-criteria analysis combined with density-based clustering. The application of density-based clustering identifies distinctive geographical clusters and determines their area and contours to compute their theoretical power generation capacity. Using density-based clustering instead of conventional clustering approaches identifies areas of irregular and required sizes without human supervision. The novelty of this study is the use of Spatial Multi-Criteria Analysis (SMCA) and DBC for the identification and contour optimization of wind farms. The only other research that used DBC is applied to solar data, as Ref [6] demonstrated, which used DBC with a limited number of spatial criteria, i.e., slope and solar irradiance. It did not discuss any final site ranking. We are augmenting this study by incorporating multiple additional spatial criteria suitable for wind farm location selection.

The classical approach for wind plant site selection has three primary analysis phases. The first phase identifies the factors and constraints of the location decision. The classical factors are the energy availability, the distance from the transmission lines, the distance from road and rail networks, distance from population centers, and the terrain suitability. These factors are categorized into multiple categories of decreasing order from highly suitable to unsuitable. Similarly, the constraints or the Boolean factors dictate whether the site is appropriate. The classical constraints are the protected areas, rivers, lakes, agricultural land, airports. The second phase of the analysis assigns the factor weights using multi-criteria decision methodologies and domain expert knowledge. This step varies for different areas of interest, as different countries or areas of interest have different development limitations. The third phase of the analysis uses the input from the first two phases and generates a combined raster output with the corresponding geographical suitability scores. This approach, when applied to regional analysis, yields a macro-level understanding of the site’s suitability. The typical site selection criteria for the wind farm are in Table 1, which the consulted experts also confirmed in Ghana’s Energy Commission.

Some limitations of the above analysis are that it introduces infrastructure proximity biases at the initial phase of the analysis, limiting the site’s optimal selection. Secondly, it does not provide the flexibility of selecting the sites of the desired minimum size, which is an essential requirement of policy planning. The size of the wind power plant is associated with the available areas, impacting the generation capacity of the wind farm. Thirdly the output is not ranked, which can cause ambiguity in setting the developmental priorities. Our proposed methodology addresses these main limitations.

The proposed hybrid methodology is aggregated into four macro steps of data preparation, zone localization and evaluation, zone clustering, and zone ranking, as shown in Fig. 1. The first three steps, data preparation, zone localization and evaluation, and zone clustering, were carried out in QGIS [45] and R [46]. The FPC package [47] was used to apply density-based clustering, and for the final visualization we used, the R package factoextra [48] and ggmap [49].

Sources of the data used for the analysis

This segment presents an overviews of the multiple spatial data set used in the analysis. All the datasets are available in the spatial projection WGS 84. The national renewable energy laboratory [50] provides a vector dataset comprising roads, rails, transmission lines, administrative borders, airports, lakes, rivers, elevation data, and land use. The dataset on protected areas includes the vector format for all the national parks, forest reserves, wildlife sanctuaries, and resource reserves [51]. The dataset used to assess the spatial distribution of the population Refs. [52,53]. This dataset is in GeoTIFF format at a resolution of 3 arcs (approximately 100 m at the equator). Finally, the data set on average wind speed was extracted from [45], which provides wind resource mapping at 250 m horizontal grid spacing at 10, 50, 100, 150, and 200 m above ground level. The summary of the geographical attributes of the country and the different data resources utilized for the analysis are in Fig. A1 in the appendix.

Data Preparation

The first step is to prepare the different raster and vector datasets. Several wind speed rasters at heights of 50, 100, 150, and 200 m are combined into a single raster. Wind turbines of different generation capacities have blades of varying lengths, which dictates the height of the generator above the ground. The effective heights at which turbine blades operate range from 50 to 200 m. Our objective is to identify the most wind-intensive areas for installing the wind farm, which is why we have aggregated the wind data in a single raster. Another approach could be to limit the analysis to a specific height, but this approach requires preselection of the wind turbines, which is seldom the case in the regional analysis. This aggregated wind speed raster data is used subsequently in the site selection decision.

Subsequently, after considering the appropriate minimum buffer requirements, the Boolean factors are rasterized. These requirements are based on existing literature on wind farm site selection presented in Fig. 1. The other factors important in the siting decision are also briefly discussed. Wind speed is one of the most critical technical criteria considered during planning to develop wind farms. The higher the wind speed at a locality, the higher the potential output of the wind power plant. It is one of the criteria that usually receive a high score from experts in the sector during such analysis [54,55].

Land-use type is also a critical assessment factor before investment in most energy projects. Experts recommend the selection of barren land for the development of wind energy projects. Areas with shorter vegetation are most suitable than areas with tall vegetation as tall vegetation can affect the wind speed by introducing turbulence to the flow of air [18,44,56]. Our analysis has only considered the open wildland or shorter vegetation areas and the arid regions to install the wind farms and excluded the agricultural and forest regions. Similarly, we excluded

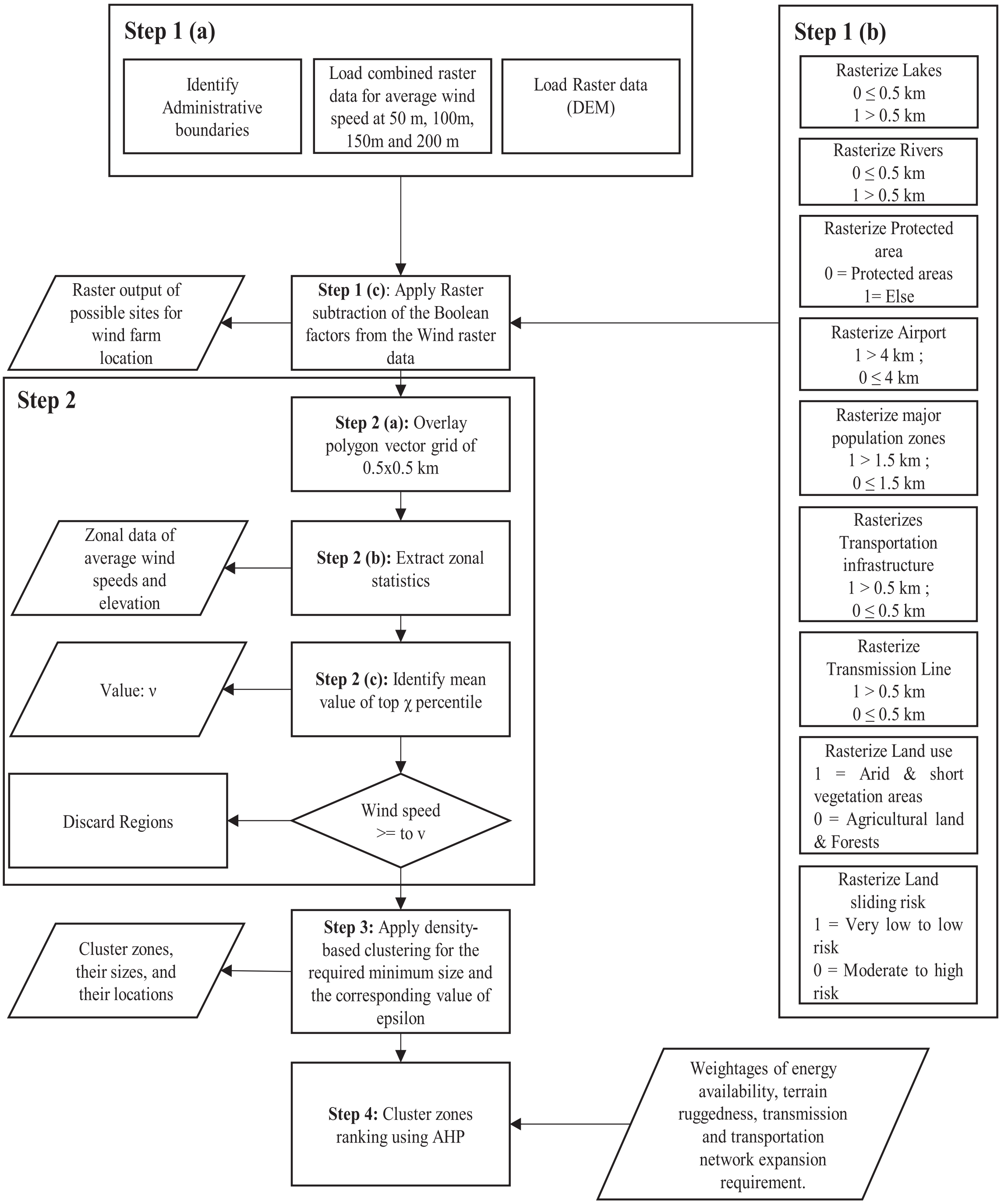

Fig. 1. Process flow diagram of the hybrid model for wind farm location selection.

protected areas, which comprise nature reserves, protected forests. These areas are essential to maintain the natural wildlife. We also excluded Lakes and rivers from the analysis by maintaining a minimum buffer of 0.5 km from the water bodies.

The construction cost of a wind farm may likely increase at areas with steeper slopes since such areas will require more earth movements and grading than a gentle slope. Additionally, it may hinder the accessibility of trucks and cranes transporting the wind turbines to the sites. Experts believe wind farm locations should be in areas with a maximum slope threshold ranging from 10% to 45% [44]. Ghana's terrain is relatively smooth, with the highest elevation of 830 m above sea level but is prone to land-sliding issues due to rainfall and minor earthquake. For this, we incorporated the data set of land-sliding and have excluded the regions of high risk from land-sliding [57].

Proximity to infrastructure, i.e., either human settlement or transportation and transmission network, is essential for the siting decision. A buffer between wind farms and residential areas reduces the potential noise and nuisance caused to residential areas. Similarly, siting wind farms near airports is discouraged as wind turbines can interfere with signals from the aviation radar [23]. For economical and accessibility reasons, wind farms should also be near existing road and transmission networks. Siting such facilities close to the available infrastructure minimizes construction costs [44,55]. However, in the context of infrastructure expansion planning and developing new renewable energy resources, these constraints must be considered later when considering the prioritization of the zone development.

After the preparation of the data, the Boolean factors are subtracted from the corresponding wind speed raster using a raster subtraction

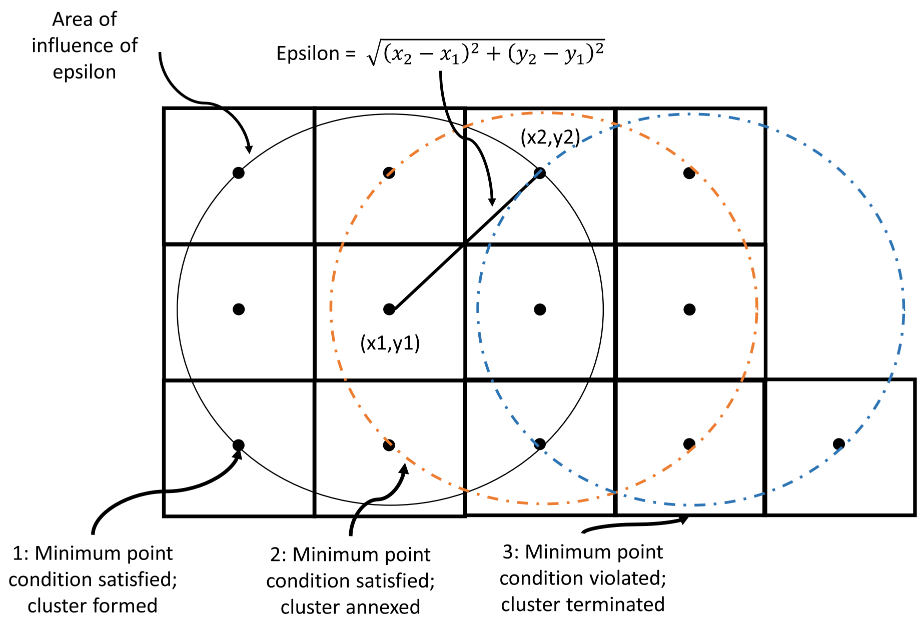

Fig. 2. Density-based clustering procedure, extracted from [6].

Table 2
Scale of relative importance [62].

The intensity of criteria *i* to *j* (a_{ij})	Explanation
1	Criteria *i* and *j* have equal importance
3	Criteria *i* is slightly more important than criteria*j*
5	Criteria *i* is moderately more important than criteria*j*
7	Criteria *i* is strongly more important than criteria*j*
9	Criteria *i* is extremely more important than criteria*j*
2,4,6,8	Intermediate values

feature of the QGIS, as shown in Step 1(c) of Fig. 1. The output from this step provides all the available areas for the site selection of the wind power plant. The next phase of the analysis will focus on evaluating the output of this step.

Zone localization and evaluation

Regional wind speed descriptive statistics are compiled on a vector grid of 0.5*0.5km^2 (Step 2a). The zonal statistics feature of the QGIS (Step 2b) generates these descriptive statistics. In the zone evaluation phase of the methodology (Step 2c), we make several decisions. For instance, include or exclude wind speed above a certain threshold and consider top 5%, 10% percent sites in energy availability. The latter decision is linked to the individual country's preferences to exploit its renewable resources. We are only concerned with developing the top 10 % of the wind resources of Ghana.

Zone clustering

In step 3, we conducted the density-based clustering on the evaluated regions of the previous step. The advantages of density-based clustering are that this technique identifies dense clusters of irregular shapes. The other clustering approaches, i.e., k-means [58] and k-medoid [59], usually make clusters around central points; concentric clusters are rare in geographical context because of the terrain irregularities. The other advantage is identifying minimum-sized clusters by adjusting the values of epsilon and minimum size requirements. Fig. 2 shows pictorially the procedure of the density-based clustering and how the parameter of epsilon values and the corresponding size of the cluster impact the clustering procedure. As the clustering proceeds, the minimum size requirement is checked, in this case 9, and the cluster is formed. The clustering procedure is terminated if the required minimum size required is not satisfied. The output from this step is the formation of clusters of the minimum size which satisfy the constraints defined in the zone evaluation phase.

Zone ranking

The final stage of the analysis includes zone ranking (Step 4). In this step, the individual clusters are ranked on multiple criteria using the AHP. This step produces a ranked list of locations for wind farm development and provides a road map for future transmission network development. The criteria considered in this step are the energy availability of the cluster, terrain ruggedness of the sites, which is a proxy of the land development requirement, its proximity to transmission lines, and its proximity to transportation networks. As the energy-dense cluster are already identified using density-based clustering, this proximity to the transportation and transmission network is the additional infrastructure required to develop these resources.

For economical and accessibility reasons, wind farms should be sited near existing road networks. Siting such facilities close to roads minimize construction costs by allowing easy transportation of equipment for construction and maintenance purposes. These are classical site selection narratives. We are deliberately applying these evaluation criteria in the last stage of analysis to remove the infrastructure proximity biases, which are already discussed in the initial part of the paper.

Analytic hierarchy process (AHP) methodology

The AHP methodology falls within the larger class of MCDM [60,61]. Each criterion in an AHP is pairwise compared, and the relative values are evaluated according to the level of importance indicated in Table 2. A pairwise comparison matrix is then formed using the input from the relevant domain experts. The domain experts comprise experts Ghana's Energy Commission, Ministry of Energy, and some other well-established personalities in the country's energy sector. To apply the AHP following steps were taken to arrive at the weights for each factor [62]:

Table 3
Random Index [66].

n	1	2	3	4	5	6	7	8	9	10
RI	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

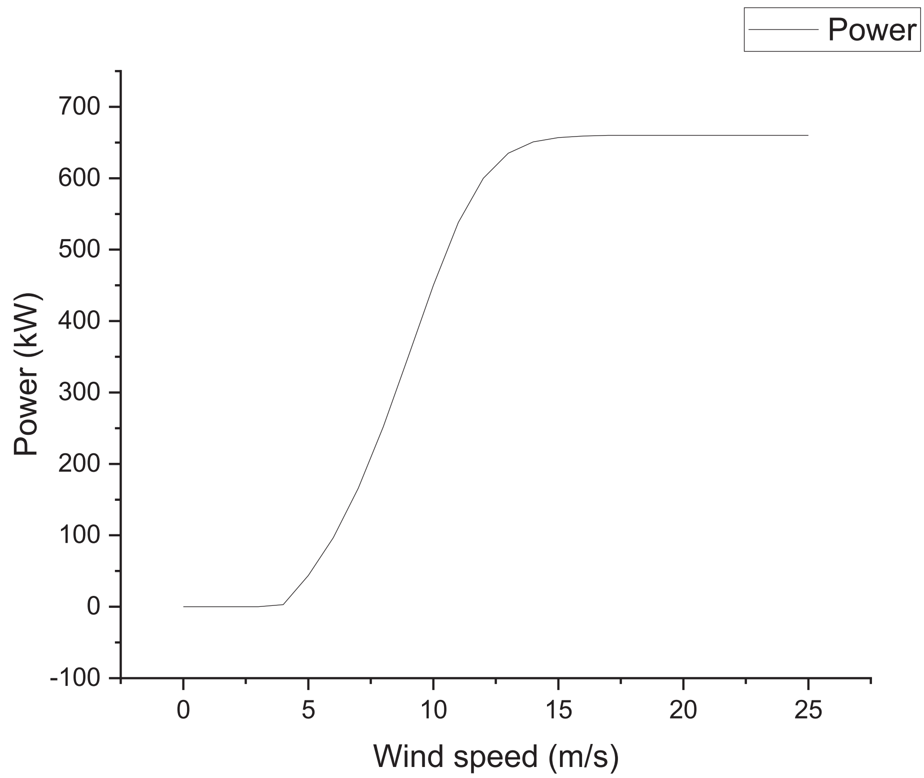

Fig. 3. Power curve for the selected wind turbine.

Let M be the pairwise comparison matrix which is a square matrix $(n \times n)$, where n represents the number of sub-criteria. Every cell a_{ij} of the matrix, M denotes the comparison values between the criterion in the *ith* row relative to the criterion in the *jth* column. Where the cell $a_{ij} > 1$, then the *ith* criteria are more important than the *jth* criteria and vice versa. However, if $a_{ij} = 1$, then it shows that both criteria have the same importance.

The geometric mean method (GMM) was used to consolidate the judgements from the consulted experts, this was done through the help of the AHP template designed by Ref [63]. The priorities $\boldsymbol{p_i}$ for each participant is evaluated using the row geometric mean method (RGMM). For matrix $\boldsymbol{M} = [\boldsymbol{a_{ij}}]$, the following expressions are used to normalize and also find the geometric consistency index (GCI) [64,65]:

$$r_i = exp\left[\frac{1}{N}\sum_{j=1}^{N}\ln(a_{ij})\right] = \left(\prod_{i=1}^{N} a_{ij}\right)^{1/N} \tag{1}$$

$$p_i = \frac{r_i}{\sum_{i=1}^{N} r_i} \tag{2}$$

$$GCI = \frac{2\sum_{i<j} lna_{ij} - ln\frac{p_i}{p_j}}{(N-1)(N-2)} \tag{3}$$

The final and crucial stage of the AHP approach is computing the consistency ratio (CR) of the comparison matrix. If the obtained CR for the matrix is less or equal to 0.1 or 10% (CR $\leq$ 0.1), then the level of consistency is acceptable to be used for further analysis. However, if CR $>$ 0.1, then the degree of consistency is unsatisfactory, and there is the need to revise the experts' judgments. The random index (RI) for specific n use in calculating the CR is presented in Table 3.

Twenty-six experts were consulted for their inputs for the AHP analysis in Ghana, mainly through face-to-face interviews and emails. In the end, twenty-three responses were received, out of which three were invalid responses because they were incomplete. A total of twenty were considered valid and used to calculate the various weights used for the analysis.

Techno-economic analysis

This section of the paper assesses the techno-economic potential of wind power plants for the country's suitable clusters using the HOMER Pro software. It is a tool designed for optimizing RETs. It conducts many simulations to identify the optimum possible matching between demand and supply, which helps developers during design stages [67]. Data for the simulations is from the National Aeronautics and Space Administration (NASA). It is a monthly average wind speed of 50 m above the earth's surface for 30 years (January 1984 – Dec 2013). This analysis augments the policy support process and demonstrates the compatibility of our proposed methodology with the already mature decision tools.

Technical feasibility aspect

For the technical analysis, wind speed distribution is critical in evaluating the energy potential in a specific site. The Weibull distribution is a good fit for the estimation of wind speed and potential for candidate sites. The Weibull probability density function (PDF) can be estimated using Eq. (4) [67,68]. The Vestas V47 wind turbine was

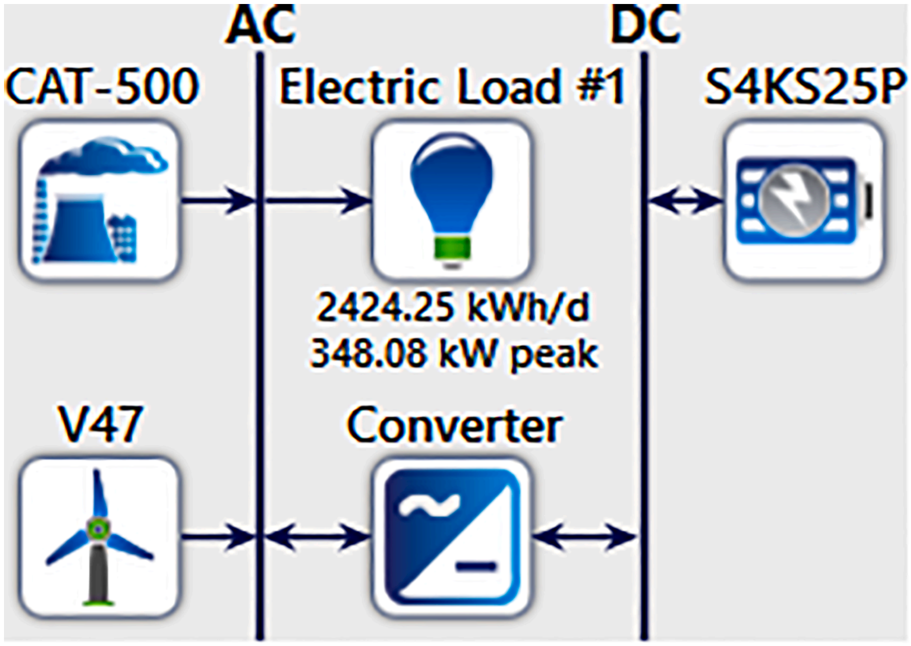

Fig. 4. Schematic of the power plant system.

selected for the analysis. This turbine has a hub height of 50 m and a lifetime of 20 years. Fig. 3 shows the power curve of the vestas wind turbine.

$$f(v) = k\Big/c\bullet(\mathrm{v/c})^{\mathrm{k}-1}\bullet\exp\Big[-\ (\mathrm{v/c})^{\mathrm{k}}\Big] \tag{4}$$

Where$k$ is the Weibull shape factor, c denotes the Weibull scale parameter, v signifies the wind speed (m/s).

The power curve of a wind turbine signifies the amount of power produced by the turbine as a function of the wind speed at a particular height. However, the power output for a particular wind turbine depends on the wind speed. Therefore, using the quadratic model as indicated in Eq. (5), the turbine's output power can be computed [67,69].

$$P_w(V_v) = \begin{cases} P_n\bullet\dfrac{v_v^2 - v_d^2}{v_n^2 - v_d^2}, V_d < V_v < V_nP_n, V_n \le V_v < V_c0, V_v \le V_d etV_v \ge V_c \end{cases} \tag{5}$$

In this case, the nominal power is denoted by P_n, the outpower is denoted by P_w, V_c signifies the cut-off speed of the wind, V_d represents the cut-in speed of the wind, and V_n also denote the rated wind speed.

The wind power can be calculated using Eq. (6) [67], but the wind speed at a particular height must be known.

$$\frac{v(z)}{v(z_a)} = \left(\frac{z}{z_a}\right)^{\alpha} \tag{6}$$

Where the mean wind speed at the desired height z which is represented by $v(z)$, α is the power index and the wind speed at the anemometer height z_a is denoted by $v(z_a)$.

The capacity factor (CF) denotes the time fraction of the operation of the wind turbine over the year on the site. It can be calculated using Eq. (7) [69].

$$CF = \frac{E_{out}}{E_r} \tag{7}$$

where E_{out} (Wh) denotes the wind turbine's annual energy generated as expressed in Eq. (8), and E_r (Wh) also represent the rated energy as expressed in Eq. (9).

$$E_{out} = \Delta T\bullet\int_0^{\infty} \mathrm{P_w}\bullet \mathrm{f(v)}\bullet\mathrm{dv} \tag{8}$$

where all parameters are as defined supra.

$$E_r = 8760\bullet\mathrm{P_r} \tag{9}$$

where the rated power is denoted by P_r (W).

Diesel generator

The electrical output power for a diesel generator plant is a function of the fuel's lower heating value (LHV), electrical efficiency, and fuel mass flow rate. To compute the output electricity from the diesel generator, we used Eq. (10) [70]:

$$P_{DG,e} = m_{fuel} \times LHV_{fuel} \times \eta_{DG,e} \tag{10}$$

where the diesel generator's electrical efficiency is represented by $\eta_{DG,e}$ and the mass flow rate of the fuel is represented by m_{fuel}. The fuel cost used for the assessment was 0.993 $/L, which is the current cost of fuel in Ghana [67].

The heat output of the diesel generator, however, is contingent on the heat recovery ratio ($\eta_{DG,hr}$) as indicated in Eq. (11).

$$P_{DG,h} = m_{fuel} \times LHV_{fuel} \times \left(1 - \eta_{DG,e}\right) \times \eta_{DG,hr} \tag{11}$$

Battery storage

The battery system is integrated into the system to store excess generated electricity from the WPP. This stored power is relied on during deficiency in electricity from the WPP unit [67]. The schematic for the simulated power plant is in Fig. 4.

Economic feasibility aspect

We can use several financial indicators for the energy sector economic viability assessment. Some of these include the levelized cost of electricity (LCOE), Net Present Cost (NPC), Equity Payback Period (EPP), Internal Rate of Return (IRR), and Simple Payback Period (SPP) [68]. This study assessed the bankability of WPP at the various identified

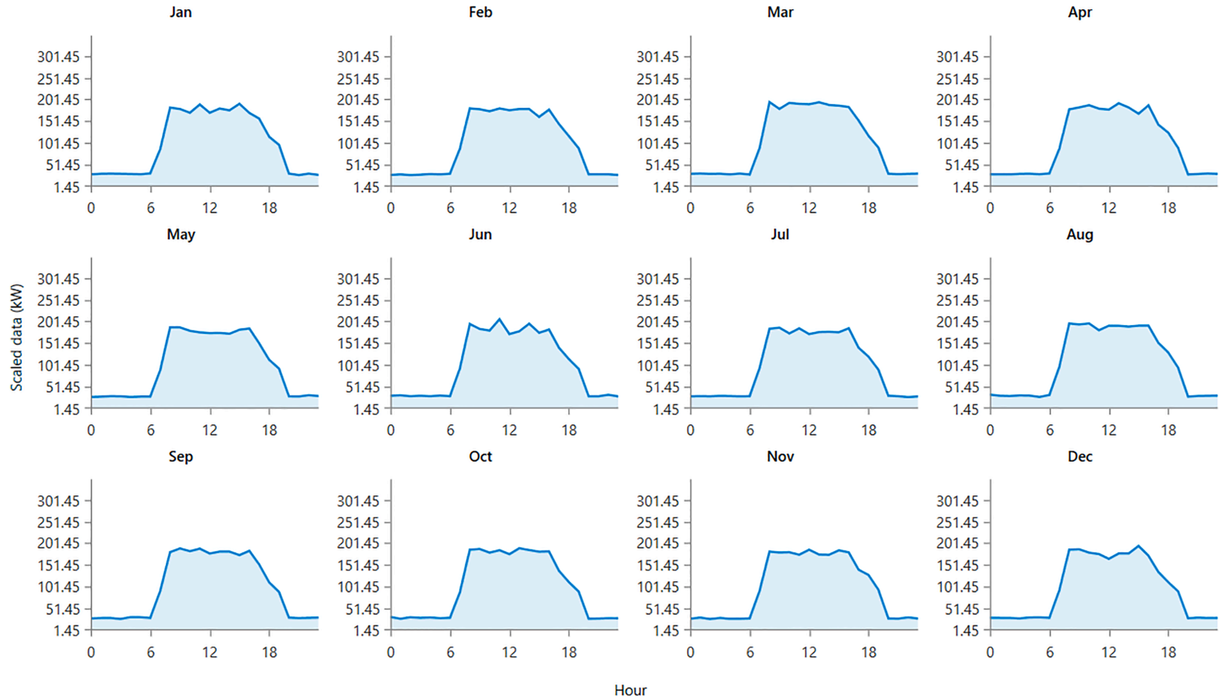

Fig. 5. Monthly load profile used for the study.

sites in the country by assessing these indicators and comparing them with each other to give both policymakers and other stakeholders a fair idea of the site's viability.

Levelized Cost of Electricity

The LCOE is usually employed to calculate the unit cost of electricity produced and is the ratio of the present value of the entire cost incurred during the power plant's lifetime and the total energy that is extractable during that period. It can be calculated using Eq. (12) [71,72].

$$Min_{LCOE}\left(\frac{\$}{kWh}\right) = \frac{NPC_T{*}CRF(\vartheta,\partial)}{\sum E_{gen}(t)} \tag{12}$$

where the minimum LCOE is represented by Min_{LCOE}, the capital recovery factor is also represented by CRF, NPC_T denotes the total NPC. The annual discount rate and the life of the plant are represented by ϑ and ∂, respectively. The energy produced by the power plant in the tth hour is denoted by $E_{gen}(t)$. The CRF converts the overall NPC to the yearly capital cost using Eq. (13).

$$CRF = \frac{\vartheta(1+\vartheta)\partial}{(1+\vartheta)\partial - 1} \tag{13}$$

Also, the NPC comprises the operations and maintenance cost (O&M) and the present value of the initial investment cost (*IIC*). Expressed mathematically, as indicated in Eq. (14) [71].

$$NPC_T = IIC + \frac{\sum_{t=1}^{N}(O\&M)_t}{(1+i)^t} \tag{14}$$

where the discount factor is represented by i, and $(O\&M)_t$ represents the operations and management cost for the tth year, and the N denotes the total number of years.

Simple payback period

SPP is commonly determined taking the ratio of the total initial investment costs and the yearly cash flow. Mathematically expressed as indicated in Eq. (15) [73].

$$SPP = \frac{C - I}{(C_e + C_s) - (C_f + C_{O\&M} + C_c)} \tag{15}$$

In this instance, grants and incentives are represented by I, the annual life cycle savings is also represented by C_s, the yearly revenue from the electricity generation is denoted by C_e, expenditure for fuel is represented by C_f,C_c denote annual debt cost, $C_{O\&M}$ signifies the annual operation and maintenance cost, and C is the total initial cost.

Internal rate of return

It is a metric employed in the financial analysis to evaluate the bankability of a potential investment. The IRR is the discount rate, which makes the net present value (NPV) for all cash flows equivalent to zero in a discounted cash flow analysis [4,74]. It is calculated using Eq. (16).

$$\sum_{i=0}^{N}\frac{C_i}{(1+IRR)^i} \tag{16}$$

where C_i represents the cash flow for i number of years, and N denotes the project lifetime and years.

Net present value

The NPV is defined as the' present value of the difference amount among the cash inflows and outflows over the project's lifetime and can is indicated in Eq. (17) [3,68,75].

$$NPV = \sum_{i=0}^{N}\frac{\widehat{C}_i}{(1+r)^i} \tag{17}$$

where the discount rate is represented by r, the number of years is represented by i, and $\widehat{C}_i$ denotes the after-tax cash flow.

Capital cost 1500 $/kW, the replacement cost 1500 $/kW, and O&M cost 0.03 $/year as used in [67,76] were adopted for the simulations for the WPP. The load profile for the simulation is represented in Fig. 5.

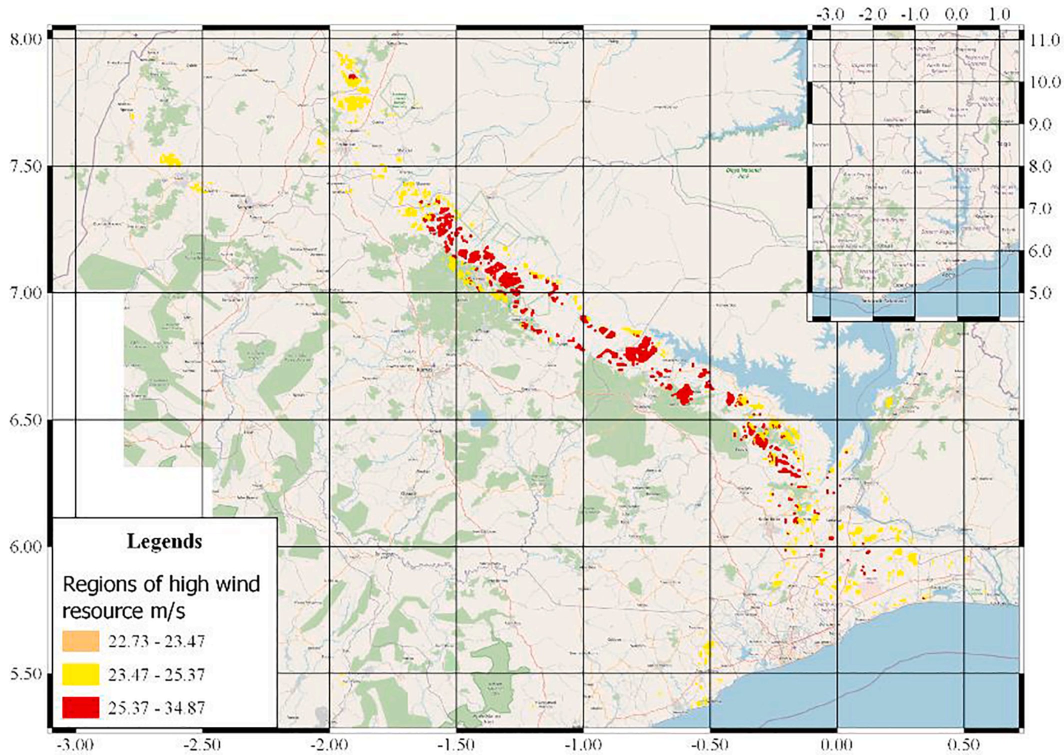

Fig. 6. The top 10 percent of the wind energy sites.

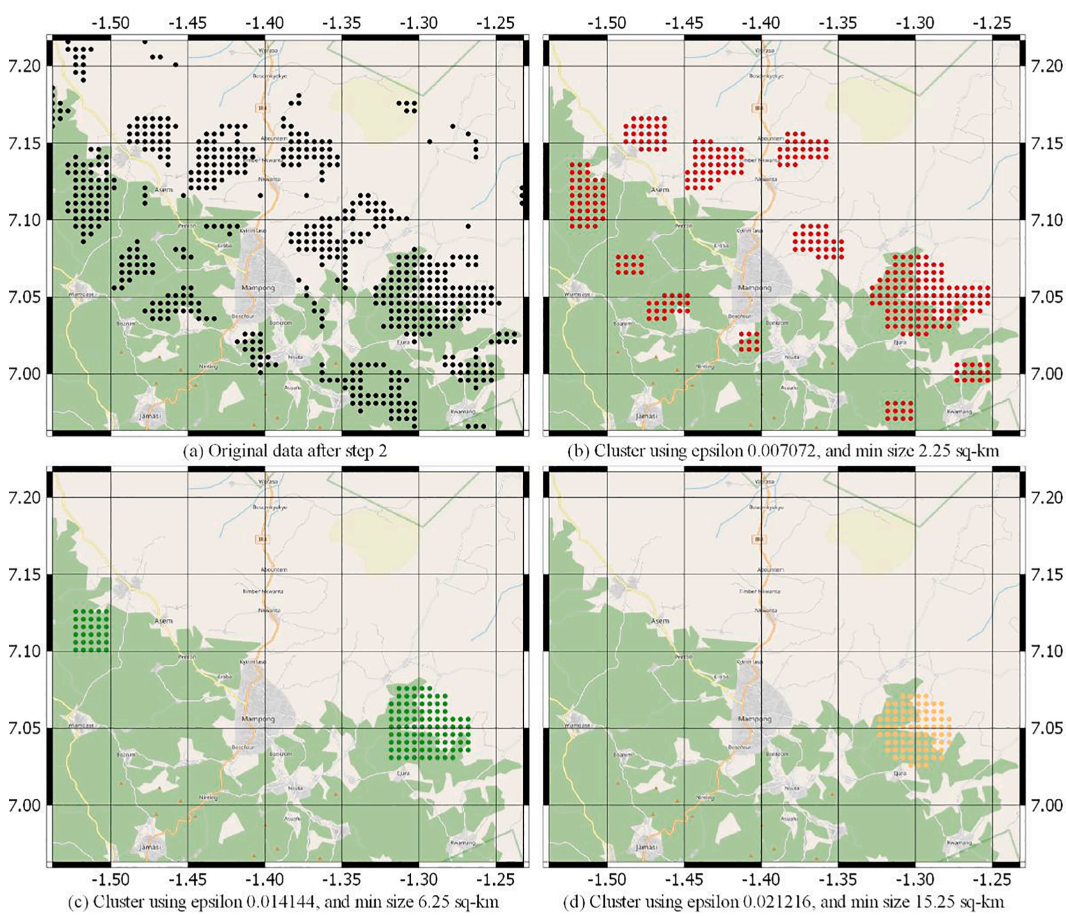

Fig. 7. Cluster contours formation by varying the DBC parameters.

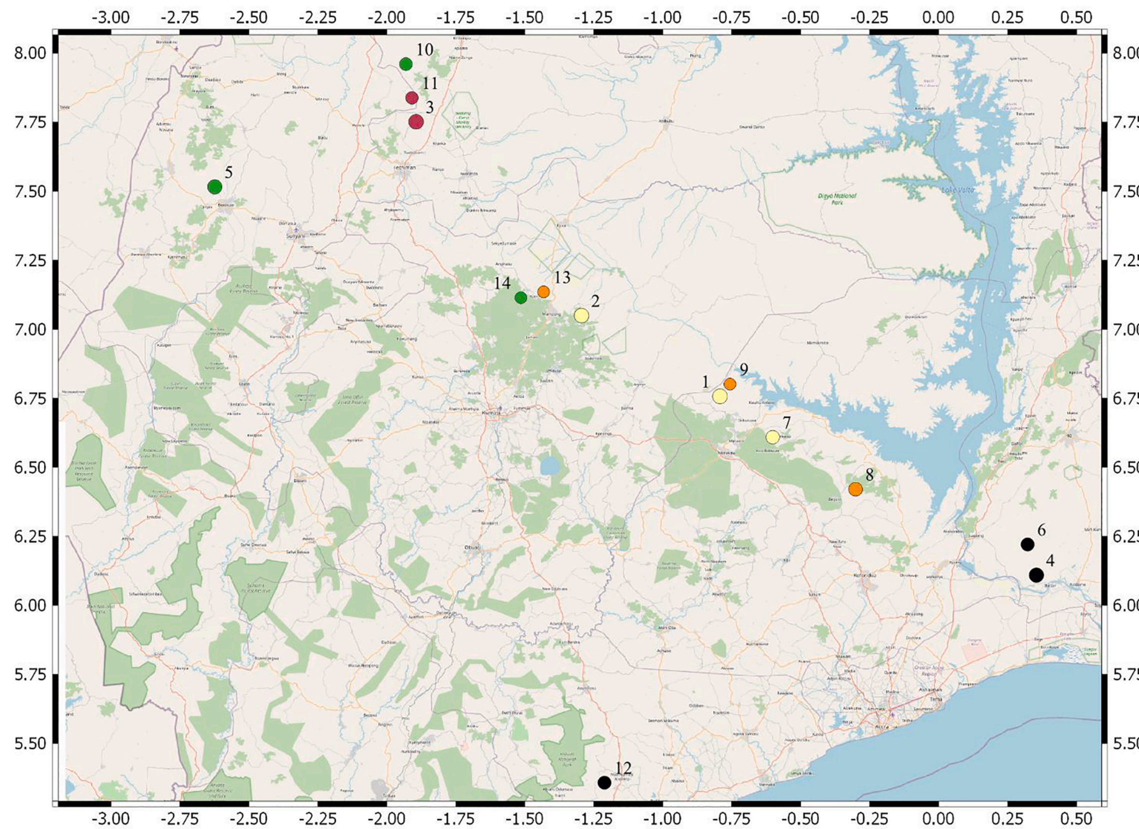

Fig. 8. Sites ranking for development of wind farm, Ghana.

Results and discussions

This section presents the results attained using the proposed methodology. It also includes a techno-economic analysis of potential hybrid energy systems, i.e., Wind/DG/Battery/Converter for the identified sites.

The outcome of the implementation of DBC

The first step of the methodology excludes all the unfavorable locations and provides only those sites that meet the minimum demography, environmental, and infrastructure variables. Fig. A1 in the appendix shows the spatial representation of all these variables. Step 2 evaluates the available sites for their average wind speeds using descriptive statistics, the top 10 percent of the sites available in Ghana have average combined wind speeds of 22.07 m/s. These statistics give the regions which are the most suitable for installing wind farms. Fig. 6 presents these suitable sites and represents approximately 1936 km^2, and provides the macro understanding of the availability of the resources but does not provide dense clusters available for the siting of the wind farm. Most of these sites are concentrated in the south to the central part of the country, with relatively small regions of high wind availability present in the country's northern section.

Step 3 of the analysis is about the application of density-based clustering to detect geographically dense clusters of the required sizes. This application requires selecting an appropriate value of the parameters of the DBC, i.e., the minimum size of the cluster and the corresponding epsilon values, refer to Fig. 2. The minimum size of the cluster depends on the rated generation capacity required from the wind farm and is an exogenous policy decision. In the case of Ghana, we have tested multiple scenarios by varying the minimum size for the required wind farm size ranging from 2.25, 6.25, and 15.25 km^2 with the corresponding epsilon values ranging from 0.007072, 0.014144, and 0.021216, respectively. By progressively increasing the parameters of the DBC, the larger available dense clusters are identified and localized. The cluster pattern formation analysis reveals that the best contour optimization can be achieved using the DBC parameters of epsilon 0.007072 and the minimum size of 2.25 km^2. These parameters of the DBC not only provide the smallest cluster but also identified the most significant clusters of complex contours and identified 117 distinct clusters ranging from size 2.25 km^2 to the largest cluster of 32 km^2. The comparison is presented in Fig. 7 between the corresponding output of the DBC at a specified location using the different parameters. Using the original data after step 2 of the analysis, DBC is applied by varying the parameters, which formed progressively larger dense clusters.

The incorporation of DBC provides additional flexibility for the policymakers regarding the rated generation required from the wind farm. It provides the exact location and the theoretical generation capacity expected from the wind farm. It also provides the approximate contours of the wind farm, optimizing its generation capacity. These characteristics are not possible by any other methodology used in the spatial multi-criteria analysis. Our proposed process has this inherent contour optimization that helps maximize the wind farm's generation capacities. This methodology is suitable for power plants in which the area requirement are high, i.e., solar, wind, and tidal power plants. The proposed methodology additionally identifies and optimizes the location selection process from the regional analysis. The other methodologies only provide suitable regions that require an additional survey to determine the optimal contours for the wind farm. The next phase of the analysis ranks these identified clusters based on the cluster's energy availability, proximity to the road network, terrain ruggedness, and proximity to the transmission network.

Ranking of candidate clusters

Based on the DBC, nationally, 14 dense clusters of high wind energy

Table 4
Ranking and the site characteristics of the wind clusters.

Priority	Size in km^2	Mean TRI	Combined average mean speed (m/s)	Distance from the road (km)	Distance from the transmission (km)	Average wind speed (m/s)			
						50 m	100 m	150 m	200 m
1	31.75	1.62	31.66	4.70	10.00	6.40	7.63	8.59	9.04
2	28.5	1.68	29.56	4.00	4.50	6.10	7.30	7.96	8.19
3	32	1.14	23.75	3.00	10.00	4.71	5.78	6.52	6.74
4	29	0.92	22.55	5.00	13.00	4.26	5.14	6.15	7.01
5	26.25	1.59	23.38	3.50	13.50	4.62	5.55	6.33	6.88
6	17.25	0.96	22.52	4.00	5.70	4.29	5.17	6.11	6.95
7	18.5	2.05	29.09	4.00	4.00	6.08	7.24	7.80	7.95
8	21.25	2.14	25.88	3.20	5.00	5.33	6.43	7.02	7.10
9	10	0.47	28.09	7.00	11.00	5.59	6.71	7.59	8.22
10	14.5	1.38	22.88	2.50	9.00	4.44	5.46	6.29	6.69
11	13	1.07	24.91	3.90	13.50	5.11	6.12	6.73	6.94
12	17	1.97	22.69	3.00	7.00	4.43	5.31	6.17	6.78
13	10.25	1.57	28.40	2.00	9.50	5.78	6.86	7.61	8.14
14	10	2.39	23.68	2.30	16.00	4.66	5.61	6.40	7.01

Table 5
Specific site locations for the various identified clusters.

Identification	Clusters	x-centroid	y-centroid
A	127	−0.792	6.757
B	114	−1.294	7.050
C	128	−1.893	7.752
D	116	0.355	6.109
E	105	−2.624	7.516
F	69	0.322	6.221
G	74	−0.601	6.609
H	85	−0.301	6.421
I	40	−0.756	6.801
J	58	−1.931	7.961
K	52	−1.909	7.839
L	68	−1.211	5.357
M	41	−1.432	7.136
N	40	−1.514	7.115

were identified along with their contours to optimize for the energy output. Together these 14 clusters represent approximately 280 km^2 and can be utilized to generate renewable energy using wind. The application of the AHP gives a ranked list of clusters for the development of wind farms while simultaneously providing policy input for future transmission expansion planning. The ranking is based on the cluster's energy availability, proximity to the road network, terrain ruggedness, and proximity to the transmission network. Each country has unique infrastructure development capabilities, and the relative importance of these infrastructures in the case of Ghana are identified using AHP, the results of which are presented in Tables A1 to A3 in the appendix. The consistency ratio obtained for the AHP matrix indicated in Table A3 was acceptable; hence, the decision matrix can be used for further analysis.

An important observation is that the most prominent sites are within relative proximity of the national transmission network and the transportation network. Interestingly the highly ranked sites are not present near the coast but are present in the central part of the country near

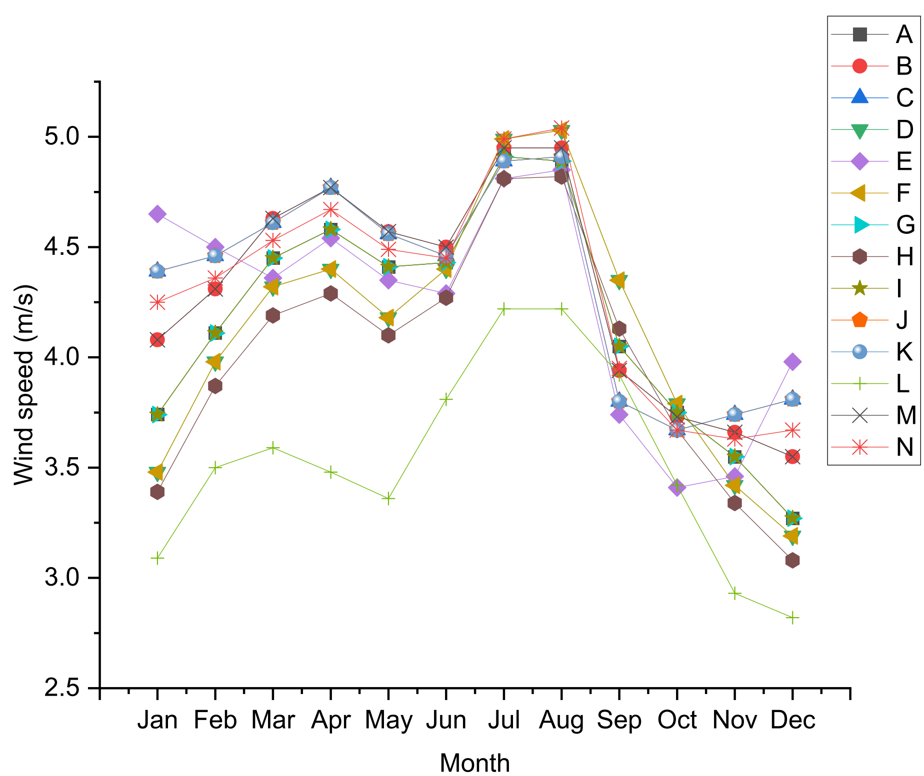

Fig. 9. Wind speed characteristics at the various clusters at 50 m (data obtained from NASA).

Table 6
Sensitivity analysis for different scenarios.

Sensitivity scenario	Technical	Economic		
	Energy availability (i.e., wind speed)	Transmission proximity	Terrain ruggedness	Transportation proximity
Current AHP weights	0.570	0.110	0.254	0.065
Equal weights	0.500	0.166	0.166	0.166
Higher economic weights	0.429	0.190	0.190	0.190

Mampong and Techiman. Fig. 8 and the corresponding tabular data on the wind energy availability are presented in Table 4. The corresponding location data of the sites are present in Fig. 8 and Table 5, respectively. The average size of the clusters is around 19 km^2, with a maximum size of up to 32 km^2.

The various clusters presented in Table 5 are also assigned identification letters to make it easier for reference for identification purposes. The exact sizes of the various clusters in square kilometers have already been calculated and presented in Table 4. Fig. 9 shows the monthly average wind speed characteristics for the various sites.

Sensitivity analysis on the AHP

In order to understand the possible effect of different views from experts in the sector, a sensitivity analysis was conducted on the AHP to assesses the effect of different weighting options on the ranking. Such analysis helps stakeholders to appreciate the impact of other factors on the final site selection. Three sensitivity scenarios are developed to check the consistency of the cluster ranking. The current AHP scenarios give a weight of 0.570 to energy availability (i.e., technical parameter) and 0.429 to economic factors. We developed two more scenarios; in the first scenario, we assigned equal weights to the technical and economic factors, i.e., 50 percent technical and 50 percent economic. We reversed the current AHP weights by giving 0.43 to technical and 0.570 to economic factors in the second scenario, as shown in Table 6. Under these two additional scenarios, we checked the change in the ranking of the identified clusters.

According to the results, as indicated in Fig. 10, the effect of the two other scenarios had a significant effect on the final rankings of the various sites. For instance, clusters A, B, C, D remained unchanged under equal weights scenario. However, the same cannot be observed about the higher economic weights scenario because all the priority areas had their positions changed under that scenario. For example, under the base case scenario where experts gave more prominence to technical parameters, the priority is cluster A, which is 31.75 km^2 with a distance to road and transmission of 4.7 km and 10 km. However, under the case where economic factors are given higher importance, the priority area will be cluster B, a total area of 28.5 km^2 with a distance to road and transmission network of 4 km and 4.5 km. Thus, the results from the various scenarios have shown the significant effect of both technical and economic factors on the final site selection.

Techno-economic analysis

The results of the technical analysis for the various WPP at the various sites are presented in this section. The HOMER software was employed to evaluate different scenarios of WPP relative to their techno-economic potentials.

Technical analysis

The economics of a power plant is greatly affected by its capacity factor and the annual electricity generation because these factors are vital input variables for assessing the economics of the system. Results from the simulations, as indicated in Table 7, show that cluster C located

Table 7
Electrical output and fuel consumption for the various sites.

Cluster	Generator output, kW h/year	WPP output, kW h/year	Excess electricity, kW h/year	Renewable fraction, %	Total fuel consumed, L
A	209,970	917,672	141,297	76.3	62,501
B	209,970	917,672	141,297	76.3	62,501
C	165,927	1,038,105	212,944	81.2	49,435
D	237,686	890,589	146,854	73.1	70,761
E	185,467	957,020	153,754	79.0	54,938
F	209,970	917,672	141,297	76.3	62,501
G	209,970	917,672	141,297	76.3	62,501
H	274,353	812,408	108,687	69.0	81,496
I	209,970	917,672	141,297	76.3	62,501
J	274,353	812,408	108,687	69.0	81,496
K	274,353	812,408	108,687	69.0	81,496
L	451,174	552,444	48,145	49.0	134,055
M	170,198	992,261	172,611	80.2	50,558
N	166,376	994,646	170,724	81.2	49,435

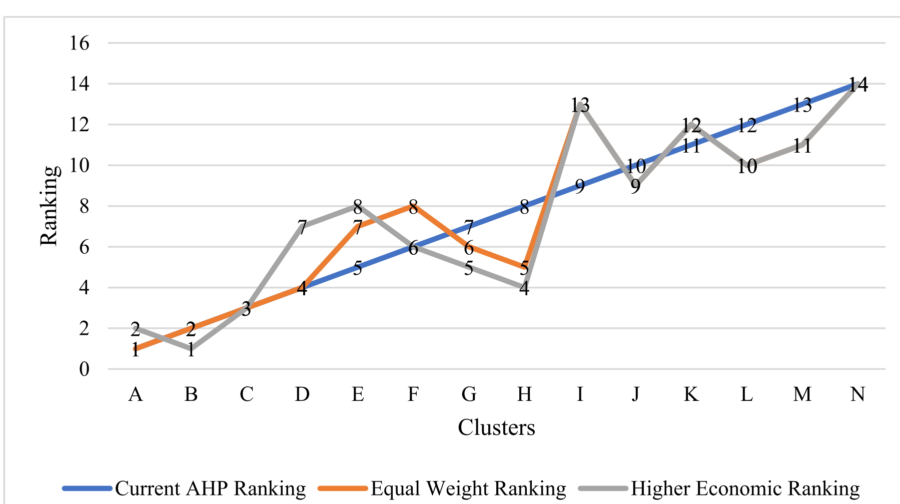

Fig. 10. Sensitivity analysis on the AHP.

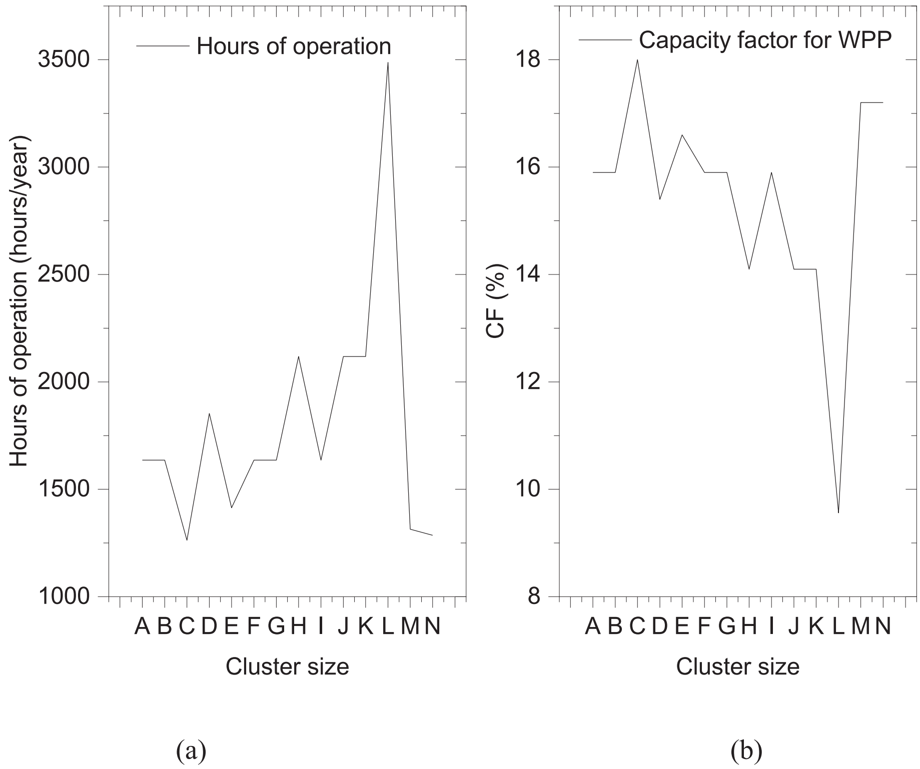

Fig. 11. Operation hours (a) for diesel generator and CF (b) for WPP.

Table 8
Economic results for the various sites.

Site	NPC, $	LCOE, $/kWh (DG/WPP/Battery system)	OC, $	LCOE, $/kW h (WPP, only)
A	2,084,902	0.22	76,385.66	0.11
B	2,084,902	0.22	76,385.66	0.11
C	1,983,466	0.21	63,528.74	0.10
D	2,153,978	0.23	84,335.23	0.11
E	2,023,936	0.21	69,099.23	0.11
F	2,084,902	0.22	76,385.66	0.11
G	1,964,796	0.22	76,385.66	0.11
H	2,268,500	0.24	95,025.09	0.13
I	2,084,902	0.22	76,385.66	0.11
J	2,268,500	0.24	95,025.09	0.13
K	2,268,500	0.24	95,025.09	0.13
L	2,753,957	0.29	146,254.50	0.18
M	1,967,386	0.21	64,618.32	0.10
N	1,964,796	0.21	63,633.19	0.10

Table 9
Economic indicators for the DG/WPP/Battery system for all sites.

Site	Present worth, $	Annual worth, $/year	ROI, %	IRR, %	SPP, years	Discounted payback, %
A	1,653,340	152,813	17.4	22.0	4.45	5.69
B	1,653,340	152,813	17.4	22.0	4.45	5.69
C	1,754,775	162,188	17.7	22.4	4.37	5.57
D	1,584,263	146,428	17.0	21.6	4.52	5.80
E	1,714,306	158,448	17.6	22.3	4.39	5.60
F	1,653,340	152,813	17.4	22.0	4.45	5.69
G	1,653,340	152,813	17.4	22.0	4.45	5.69
H	1,469,742	135,844	16.2	20.7	4.70	6.09
I	1,653,340	152,813	17.4	22.0	4.45	5.69
J	1,469,742	135,844	16.2	20.7	4.70	6.09
K	1,469,742	135,844	16.2	20.7	4.70	6.09
L	984,284	90,974	13.0	17.2	5.53	7.54
M	1,770,855	163,674	18.1	22.8	4.30	5.46
N	1,773,445	163,914	18.1	22.7	4.31	5.48

around the Techiman enclave has the highest electricity generation from the WPP. The lowest electricity output for the WPP plant was recorded at cluster L, an area around Abura Asebu Kwamankese in the Central region. It, therefore, suggests that the diesel generator operated longer hours in cluster L, which has implications not only on the economics of the power plant but also on the level of emissions from the plant into the environment. The hours of operation for the diesel generator and the CF at the various cluster sites are represented in Fig. 11. It is clear from the results that the higher the CF of the WPP, the lesser the operational hours of the diesel generator. It, therefore, suggests that increasing the wind speed at a particular site could lead to a reduction in the consumed fuel. Therefore, the turbine's hub height can be increased from the current 50 m used for the analysis to experience higher wind speeds since, as indicated in Table 4, the wind speed is higher at higher heights. The highest electricity output for the WPP occurs in July and August, with the least occurring in the last three months of the year for all sites.

Economic analysis

The financial viability of any project is part of the essential parameters that any investor will want to look at before making an investment decision. The economic analysis identifies specific financial indicators such as the LCOE, IRR, NPC, OC, SPP, and ROI. A comparative data for the various identified candidate sites is presented in Table 8. The LCOE of all sites fundamentally falls within a range of 0.20–0.30 $/kW h. The LCOE for a standalone WPP without the diesel generator indicated in Table 8 shows a reduced LCOE compared to the hybrid DG/WPP/Battery/Converter system. The reduction in LCOEs is attributed to the fact that there was no fuel cost in that system. The current cost of electricity in Ghana ranges between 12 and 17 ¢/kW h for the residential sector, 15–25 ¢/kW h for the non-residential sector, and 16 – 22 ¢/kWh for the industrial sector (i.e., particular load tariff system for industrial energy use purposes; supply voltages HV-High Voltage (33,000 V); MV–Medium Voltage (11,000 V); and LV–Low Voltage [77]. The DG/WPP/

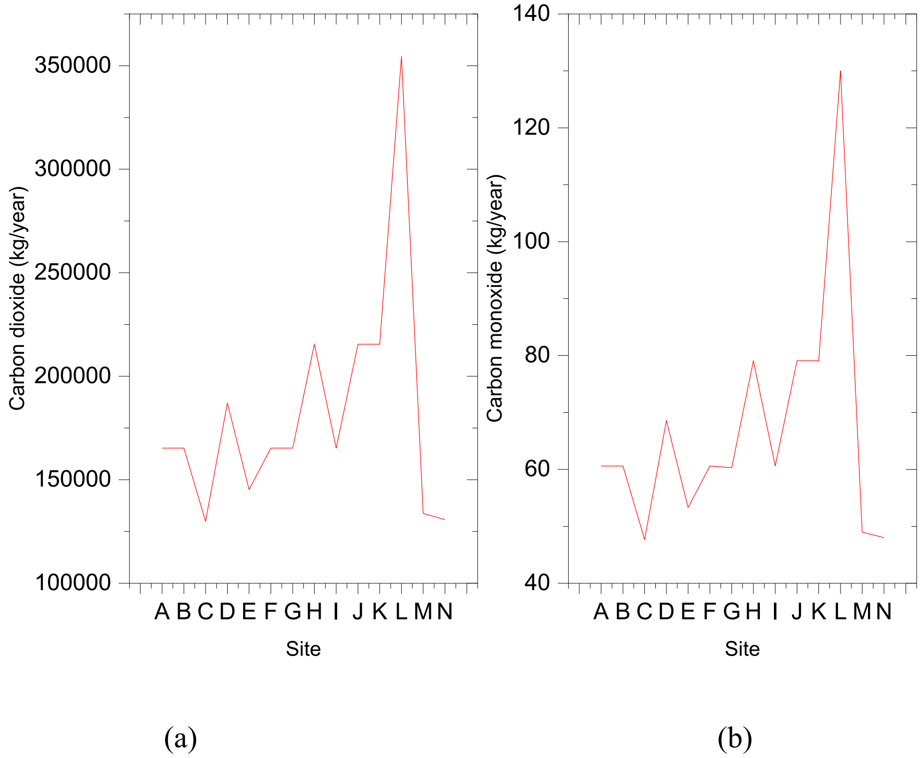

Fig. 12. Carbon dioxide (a) and Carbon monoxide (b) emissions for each site.

Table 10
Other studies for comparative analysis.

Hybrid system	Year	Peak load, kW	LCOE, $/kWh	NPC, $	Refs.
WT/DG/BT	2016		0.260	57,320	[82]
Wind/Diesel	2017	8.70	0.431	160,626	[83]
Wind/Diesel/Battery	2017	7.00	0.198	76,901	[84]
Wind/Diesel/battery	2020	1400.44	0.471	7,735,646	[85]
Wind/DG/Battery	2020	407.71	0.396	8,966,700	[67]
PV/WPP/Biogas Generator/Fuel cell	2020	55.47	0.164–0.233		[86]
Grid/diesel/PV/battery	2020		0.042	102,290	[87]
Biogas generator/PV/DG/Battery/Inverters	2017	44.41	0.280	612,280	[88]
PV/Wind/DG/Battery/Converter	2020	49.07	0.226	351,223	[89]
PV/wind/Diesel/Battery	2019		0.110	1,010,000	[90]
PV/Diesel/Battery	2019		0.396	4,909,206	[91]

Battery/Converter system will be too expensive for the residential sector if implemented in its current form without any subsidies to the customer or incentives to the investor from the government. It will, however, fall within the tariff structure for the non-residential and industrial sectors if implemented under the conditions used for the study. However, the WPP recorded LCOEs fall below that of the DG/WPP/Battery system, which will present an affordable tariff structure for all consumers if implemented under the conditions used for the current study. The tariffs for the non-residential and industrial sectors can be reduced by almost 50% if the upper limit of the current range of tariff structure is considered.

Other fundamental economic indicators during the analysis of such projects are IRR and the return on investment (ROI). ROI is a traditional tool employed mainly in the private sector to assess and compare investments and projects. Most simply, it is defined as the ratio of the net earnings accrued from a particular project and the project costs. Currently, also used in the public sector for several evaluations [78]. ROI gives information on the level at which the amount of money invested in a particular project returns as profit or loss. In other words, it allows for the assessment of investment efficiency [79]. The payback period for projects also evaluates projects' profitability and is considered in this study. The economic indicators for the various sites are presented in Table 9. An economically viable project must have its IRR to be more than the discount rate [68]. Considering the 10% discount rate used for the analysis in this study and comparing same to the various IRR for all clusters sites suggests that all sites are economically viable for the studied project. Data from the table also suggests that the site with cluster M comes as the optimum site since the least SPP of 4.30 years and highest IRR 22.8% were obtained for the power plants at that site.

Table A1
Pairwise comparison matrix.

	Energy availability	Transmission network proximity	Terrain ruggedness	Transportation network proximity
Energy availability	1.00	5.00	3.00	7.00
Transmission network proximity	0.20	1.00	0.33	2.00
Terrain ruggedness	0.33	3.00	1.00	4.00
Transportation network proximity	0.14	0.50	0.25	1.00
Total	1.67	9.50	4.58	14.00

Environmental impact assessment

One significant challenge globally is emissions from traditional

Table A2
Normalization of matrix.

	Energy availability	Transmission network proximity	Terrain ruggedness	Transportation network proximity	CW
Energy availability	0.599	0.526	0.655	0.500	0.570
Transmission network proximity	0.120	0.105	0.072	0.143	0.110
Terrain ruggedness	0.200	0.316	0.218	0.286	0.254
Transportation network proximity	0.084	0.053	0.055	0.071	0.065
Total	1.000	1.000	1.000	1.000	1.000

Table A3
Priorities and errors.

Criteria	Priority	Rank	Error (+/−)
Energy availability	0.570	1	12.7%
Transmission network expansion	0.108	3	1.9%
Terrain ruggedness	0.253	2	5.3%
Transportation network expansion	0.066	4	1.2%

A CR of 0.02 and a GCI of 0.08 were obtained which indicate that the consistency of the judgements from the experts is acceptable.

power plants that rely on fossil fuels for electricity generation. Additionally, many health issues and hazards are also associated with these emissions. For this reason, protocols and standards exist to minimize and mitigate greenhouse gas emissions [80]. As indicated in Fig. 12, the higher the output or CF of the RE system, the lesser the GHG emissions from the power plant due to the fewer operational hours of the diesel generator; thus, the environmental GHG emissions reduction potential will be higher.

The environmental analysis puts sites C, M, and N better than the other sites. Because of the higher CF values of the WPP recorded at those sites. The worst site for the installation will be site L because it recorded the highest emissions of about 354,474 kg/year for carbon dioxide and 130 kg/year for carbon monoxide.

Validation of results

Compared to other literature on Ghana's wind energy potential, the results validate the methodology used in our study. Data from the Denmark Technical University and the World Bank Group on Ghana's wind energy resource indicated in [81] confirms our results. However, our results provide better locational and capacity assessment as the referenced studies only provide macro-level intelligence about the presence of wind resources.

For validity comparison of the techno-economic section of this study, the existing literature in Table 10 is reviewed to help readers and other stakeholders in decision-making. A close look at the reviewed literature indicates that the results obtained in this study fall within what was

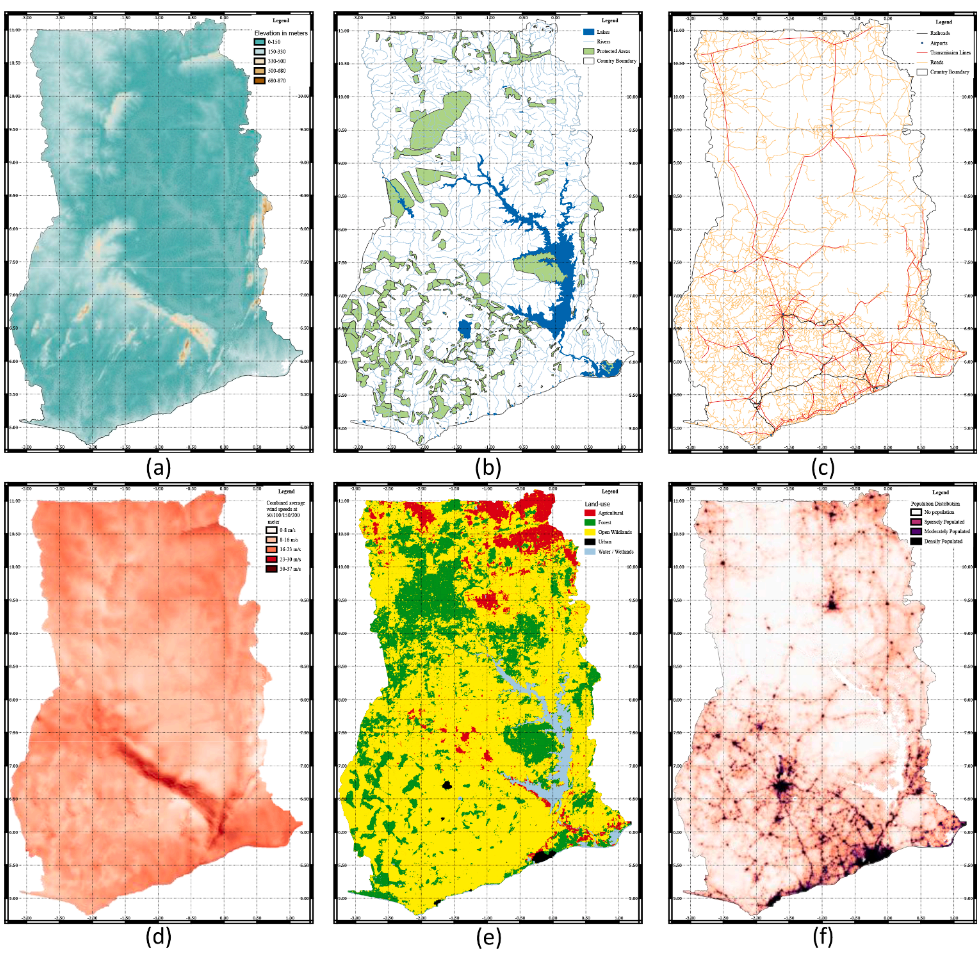

Fig. A1. Elevation (A), Protected Areas, Lakes, and Rivers (B), Road, Rail, Airports, Transmission network (C), Wind Speeds (D), Land use (E), Population Distribution (F) maps of Ghana.

obtained in other jurisdictions, notably the WT/DG/Battery systems. Those with relatively higher LCOE are mainly because of either lower CF for the renewables or integration of other components (i.e., PV or biogas generators), which increases the present value of the entire cost incurred during the project's lifetime, thereby affecting the LCOE.

Limitations of this study and future research recommendations

Despite the comprehensive nature of this study, it does have some limitations. The selection of appropriate sites for the construction of wind farms involves lots of considerations. This study considered only seven criteria in the assessment. However, other factors such as fault lines, soil type, hazardous facilities such as petrochemical installations (i.e., gas pipelines and oil) might affect the final consideration for a site selection. Therefore, this study recommends that such factors be considered in future studies to get more precise insights into potential sites. However, these limitations do not devalue the obtained results in this study since the factors considered in this study are the major factors considered for such an analysis, as evident in the reviewed literature. Another limitation encountered during this methodology is the non-availability of the density-based clustering in the GIS environment.

For the techno-economics analysis of various areas, we recommend conducting a detailed sensitivity analysis on technical and financial parameters to know their impact on the viability of such investments in future studies.

Conclusions and policy implications

The selection of an appropriate site for the development of wind farms is one of the significant implementation steps for such projects. This paper proposed a new approach for selecting the optimal sites for developing wind farms in emerging economies, different from the existing approach, which is generally biased towards the availability of existing infrastructure. The existing methodologies are mainly favorable to developed economies with well-structured and advanced infrastructural facilities spread evenly, which is not the same for most developing countries. The spatial multi-criteria analysis and density-based clustering approach implemented was used for the first time to identify sites with high wind speed in Ghana. The results provide policy input for future transmission network expansion planning. The analysis identified 14 geographical clusters with high wind energy availability. The average size of the clusters is around 19 $km^{2,}$ with a maximum area of up to 32 km^2. All the clusters found are in relative proximity to transportation networks and the transmission network. These clusters represent the most wind energy-rich regions of Ghana. The methodology that we have proposed does not suffer from infrastructure bias. This methodology applied spatial multi-criteria analysis to eliminate problematic areas, density-based clustering to identify the dense cluster and their complexity contours. The combination of the spatial multi-criteria and the density-based clustering is helpful for other location decisions of other renewable resources, which require large areas for extracting and converting renewable energy resources, i.e., solar and tidal.

The techno-economic and environmental analysis conducted identified the least LCOE of 0.21 \$/kWh at sites C, E, M, and N. The results also suggest that cluster M has the least SPP of 4.30 years, and the highest IRR 22.8% were obtained for the power plant at that site. The environmental analysis puts sites C, M, and N better than the other sites, attributed to the higher CF values of the WPP recorded at those sites due to the high wind speed.

With the obtained results, the Ghanaian government can develop a master plan for developing the country's wind energy sector by concentrating on the top 10 percent sites identified in this study. The proposed methodology can be used in any case study to identify optimal locations for developing wind energy farms anywhere worldwide, particularly in developing countries with less-developed electricity transmission networks.

Data Availability

All required data for the study are identified in the text.

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CRediT authorship contribution statement

Fahd Amjad: Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing, Software, Data curation. **Ephraim Bonah Agyekum:** Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing, Software, Data curation. **Liaqat Ali Shah:** Writing - original draft. **Ahsan Abbas:** Writing - original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

References

[1] Sarmiento N, Belmonte S, Dellicompagni P, Franco J, Escalante K, Sarmiento J. A solar irradiation GIS as decision support tool for the Province of Salta. Argentina. Renewable Energy 2019;132:68–80. https://doi.org/10.1016/j.renene.2018.07.081.

[2] Agyekum EB. Energy poverty in energy rich Ghana: A SWOT analytical approach for the development of Ghana's renewable energy. Sustainable Energy Technol Assess 2020;40:100760. https://doi.org/10.1016/j.seta.2020.100760.

[3] Agyekum EB, Velkin VI, Hossain I. Sustainable energy: Is it nuclear or solar for African Countries? Case study on Ghana. Sustainable Energy Technol Assess 2020;37:100630. https://doi.org/10.1016/j.seta.2020.100630.

[4] Agyekum EB, Velkin VI. Optimization and techno-economic assessment of concentrated solar power (CSP) in South-Western Africa: A case study on Ghana. Sustainable Energy Technol Assess 2020;40:100763. https://doi.org/10.1016/j.seta.2020.100763.

[5] Gigović L, Pamučar D, Božanić D, Ljubojević S. Application of the GIS-DANP-MABAC multi-criteria model for selecting the location of wind farms: A case study of Vojvodina. Serbia. Renewable Energy 2017;103:501–21. https://doi.org/10.1016/j.renene.2016.11.057.

[6] Amjad F, Shah LA. Identification and assessment of sites for solar farms development using GIS and density based clustering technique- A case of Pakistan. Renewable Energy 2020;155:761–9. https://doi.org/10.1016/j.renene.2020.03.083.

[7] Pillot B, Al-Kurdi N, Gervet C, Linguet L. An integrated GIS and robust optimization framework for solar PV plant planning scenarios at utility scale. Appl Energy 2020;260:114257. https://doi.org/10.1016/j.apenergy.2019.114257.

[8] Adebayo TS, Agboola MO, Rjoub H, Adeshola I, Agyekum EB, Kumar NM. Linking Economic Growth, Urbanization, and Environmental Degradation in China: What Is the Role of Hydroelectricity Consumption? Int J Environ Res Public Health 2021;18:6975. https://doi.org/10.3390/ijerph18136975.

[9] Adebayo TS, Awosusi AA, Oladipupo SD, Agyekum EB, Jayakumar A, Kumar NM. Dominance of Fossil Fuels in Japan's National Energy Mix and Implications for Environmental Sustainability. Int J Environ Res Public Health 2021;18:7347. https://doi.org/10.3390/ijerph18147347.

[10] Olaofe ZO. Review of energy systems deployment and development of offshore wind energy resource map at the coastal regions of Africa. Energy 2018;161:1096–114. https://doi.org/10.1016/j.energy.2018.07.185.

[11] He J, Chan PW, Li Q, Lee CW. Spatiotemporal analysis of offshore wind field characteristics and energy potential in Hong Kong. Energy 2020;201:117622. https://doi.org/10.1016/j.energy.2020.117622.

[12] Nsafon BEK, Butu HM, Owolabi AB, Roh JW, Suh D, Huh J-S. Integrating multi-criteria analysis with PDCA cycle for sustainable energy planning in Africa: Application to hybrid mini-grid system in Cameroon. Sustainable Energy Technol Assess 2020;37:100628. https://doi.org/10.1016/j.seta.2020.100628.

[13] Siyal SH, Mörtberg U, Mentis D, Welsch M, Babelon I, Howells M. Wind energy assessment considering geographic and environmental restrictions in Sweden: A GIS-based approach. Energy 2015;83:447–61. https://doi.org/10.1016/j.energy.2015.02.044.

[14] Mentis D, Hermann S, Howells M, Welsch M, Siyal SH. Assessing the technical wind energy potential in Africa a GIS-based approach. Renewable Energy 2015;83:110–25. https://doi.org/10.1016/j.renene.2015.03.072.

[15] Cavazzi S, Dutton AG. An Offshore Wind Energy Geographic Information System (OWE-GIS) for assessment of the UK's offshore wind energy potential. Renewable Energy 2016;87:212–28. https://doi.org/10.1016/j.renene.2015.09.021.

[16] Feng J, Feng L, Wang J, King CW. Evaluation of the onshore wind energy potential in mainland China—Based on GIS modeling and EROI analysis. Resour Conserv Recycl 2020;152:104484. https://doi.org/10.1016/j.resconrec.2019.104484.

[17] Xu Ye, Li Ye, Zheng L, Cui L, Li S, Li W, et al. Site selection of wind farms using GIS and multi-criteria decision making method in Wafangdian. China. Energy 2020; 207:118222. https://doi.org/10.1016/j.energy.2020.118222.
[18] Ali S, Taweekun J, Techato K, Waewsak J, Gyawali S. GIS based site suitability assessment for wind and solar farms in Songkhla. Thailand. Renewable Energy 2019;132:1360–72. https://doi.org/10.1016/j.renene.2018.09.035.
[19] Anwarzai MA, Nagasaka K. Utility-scale implementable potential of wind and solar energies for Afghanistan using GIS multi-criteria decision analysis. Renew Sustain Energy Rev 2017;71:150–60. https://doi.org/10.1016/j.rser.2016.12.048.
[20] Amirinia G, Mafi S, Mazaheri S. Offshore wind resource assessment of Persian Gulf using uncertainty analysis and GIS. Renewable Energy 2017;113:915–29. https://doi.org/10.1016/j.renene.2017.06.070.
[21] Ayodele TR, Ogunjuyigbe ASO, Odigie O, Munda JL. A multi-criteria GIS based model for wind farm site selection using interval type-2 fuzzy analytic hierarchy process: The case study of Nigeria. Appl Energy 2018;228:1853–69. https://doi.org/10.1016/j.apenergy.2018.07.051.
[22] Watson JJW, Hudson MD. Regional Scale wind farm and solar farm suitability assessment using GIS-assisted multi-criteria evaluation. Landscape Urban Plann 2015;138:20–31. https://doi.org/10.1016/j.landurbplan.2015.02.001.
[23] Baseer MA, Rehman S, Meyer JP, Alam MM. GIS-based site suitability analysis for wind farm development in Saudi Arabia. Energy 2017;141:1166–76. https://doi.org/10.1016/j.energy.2017.10.016.
[24] Liang Z, Chen H, Chen S, Lin Z, Kang C. Probability-driven transmission expansion planning with high-penetration renewable power generation: A case study in northwestern China. Appl Energy 2019;255:113610. https://doi.org/10.1016/j.apenergy.2019.113610.
[25] Jadidoleslam M, Ebrahimi A, Latify MA. Probabilistic transmission expansion planning to maximize the integration of wind power. Renewable Energy 2017;114: 866–78. https://doi.org/10.1016/j.renene.2017.07.063.
[26] Veronesi F, Schito J, Grassi S, Raubal M. Automatic selection of weights for GIS-based multicriteria decision analysis: site selection of transmission towers as a case study. Appl Geogr 2017;83:78–85. https://doi.org/10.1016/j.apgeog.2017.04.001.
[27] Fernandez-Jimenez LA, Mendoza-Villena M, Zorzano-Santamaria P, Garcia-Garrido E, Lara-Santillan P, Zorzano-Alba E, et al. Site selection for new PV power plants based on their observability. Renewable Energy 2015;78:7–15. https://doi.org/10.1016/j.renene.2014.12.063.
[28] Doljak D, Stanojević G. Evaluation of natural conditions for site selection of ground-mounted photovoltaic power plants in Serbia. Energy 2017;127:291–300. https://doi.org/10.1016/j.energy.2017.03.140.
[29] Cunden TSM, Doorga J, Lollchund MR, Rughooputh SDDV. Multi-level constraints wind farms siting for a complex terrain in a tropical region using MCDM approach coupled with GIS. Energy 2020;211:118533. https://doi.org/10.1016/j.energy.2020.118533.
[30] Villacreses G, Gaona G, Martínez-Gómez J, Jijón DJ. Wind farms suitability location using geographical information system (GIS), based on multi-criteria decision making (MCDM) methods: The case of continental Ecuador. Renewable Energy 2017;109:275–86. https://doi.org/10.1016/j.renene.2017.03.041.
[31] Messaoudi D, Settou N, Negrou B, Settou B. GIS based multi-criteria decision making for solar hydrogen production sites selection in Algeria. Int J Hydrogen Energy 2019;44(60):31808–31. https://doi.org/10.1016/j.ijhydene.2019.10.099.
[32] Sánchez-Lozano JM, Henggeler Antunes C, García-Cascales MS, Dias LC. GIS-based photovoltaic solar farms site selection using ELECTRE-TRI: Evaluating the case for Torre Pacheco, Murcia. Southeast of Spain. Renewable Energy 2014;66:478–94. https://doi.org/10.1016/j.renene.2013.12.038.
[33] Konstantinos I, Georgios T, Garyfalos A. A Decision Support System methodology for selecting wind farm installation locations using AHP and TOPSIS: Case study in Eastern Macedonia and Thrace region. Greece. Energy Policy 2019;132:232–46. https://doi.org/10.1016/j.enpol.2019.05.020.
[34] Latinopoulos D, Kechagia K. A GIS-based multi-criteria evaluation for wind farm site selection. A regional scale application in Greece. Renewable Energy 2015;78: 550–60. https://doi.org/10.1016/j.renene.2015.01.041.
[35] Dhiman HS, Deb D. Fuzzy TOPSIS and fuzzy COPRAS based multi-criteria decision making for hybrid wind farms. Energy 2020;202:117755. https://doi.org/10.1016/j.energy.2020.117755.
[36] Saraswat SK, Digalwar AK, Yadav SS, Kumar G. MCDM and GIS based modelling technique for assessment of solar and wind farm locations in India. Renewable Energy 2021;169:865–84. https://doi.org/10.1016/j.renene.2021.01.056.
[37] Giamalaki M, Tsoutsos T. Sustainable siting of solar power installations in Mediterranean using a GIS/AHP approach. Renewable Energy 2019;141:64–75. https://doi.org/10.1016/j.renene.2019.03.100.
[38] Brewer J, Ames DP, Solan D, Lee R, Carlisle J. Using GIS analytics and social preference data to evaluate utility-scale solar power site suitability. Renewable Energy 2015;81:825–36. https://doi.org/10.1016/j.renene.2015.04.017.
[39] Vafaeipour M, Hashemkhani Zolfani S, Morshed Varzandeh MH, Derakhti A, Keshavarz EM. Assessment of regions priority for implementation of solar projects in Iran: New application of a hybrid multi-criteria decision making approach. Energy Convers Manage 2014;86:653–63. https://doi.org/10.1016/j.enconman.2014.05.083.
[40] Wang S, Zhang L, Fu D, Lu X, Wu T, Tong Q. Selecting photovoltaic generation sites in Tibet using remote sensing and geographic analysis. Sol Energy 2016;133: 85–93. https://doi.org/10.1016/j.solener.2016.03.069.
[41] Yahiaoui A, Benmansour K, Tadjine M. Control, analysis and optimization of hybrid PV-Diesel-Battery systems for isolated rural city in Algeria. Sol Energy 2016;137:1–10. https://doi.org/10.1016/j.solener.2016.07.050.
[42] Suresh Kumar U, Manoharan PS. Economic analysis of hybrid power systems (PV/diesel) in different climatic zones of Tamil Nadu. Energy Convers Manage 2014;80: 469–76. https://doi.org/10.1016/j.enconman.2014.01.046.
[43] Yilmaz S, Dincer F. Optimal design of hybrid PV-Diesel-Battery systems for isolated lands: A case study for Kilis. Turkey. Renewable and Sustainable Energy Reviews 2017;77:344–52. https://doi.org/10.1016/j.rser.2017.04.037.
[44] Jangid J, Bera AK, Joseph M, Singh V, Singh TP, Pradhan BK, et al. Potential zones identification for harvesting wind energy resources in desert region of India – A multi criteria evaluation approach using remote sensing and GIS. Renew Sustain Energy Rev 2016;65:1–10. https://doi.org/10.1016/j.rser.2016.06.078.
[45] Quantum GIS. Development Team.(2013). Quantum GIS geographic information system. Open Source Geospatial Foundation Project. 2013.
[46] Team CR, Team RDC, R. A language and environment for statistical computing. R Foundation for statistical computing 2013;1:12–21.
[47] C. Hennig fpc: Flexible procedures for clustering. R package version 2013 2.1-5.
[48] Kassambara A, Factoextra MF. Extract and visualize the results of multivariate data analyses. R package version 1.04. 999. Available Ao Hoop://Www Sohda Com/English/Rpkgs/Facooexora 2017.
[49] Kahle D, Wickham H. ggmap: Spatial Visualization with ggplot2. The R Journal 2013;5:144–61.
[50] NREL. RED-E Ghana n.d. https://maps.nrel.gov/rede-ghana/?aL=33LD6x%255Bv%255D%3Dt&bL=clight&cE=0&lR=0&mC=7.98851761453913%2C0.5548095703125&zL=7 (accessed November 5, 2020).
[51] Protected Planet | Ghana. Protected Planet n.d. https://www.protectedplanet.net/country/GH (accessed November 17, 2020).
[52] Linard C, Gilbert M, Snow RW, Noor AM, Tatem AJ, Schumann G-P. Population Distribution, Settlement Patterns and Accessibility across Africa in 2010. PLoS ONE 2012;7(2):e31743. https://doi.org/10.1371/journal.pone.003174310.1371/journal.pone.0031743.g00110.1371/journal.pone.0031743.g00210.1371/journal.pone.0031743.g00310.1371/journal.pone.0031743.t00110.1371/journal.pone.0031743.s00110.1371/journal.pone.0031743.s00210.1371/journal.pone.0031743.s00310.1371/journal.pone.0031743.s004.
[53] Tatem AJ. WorldPop, open data for spatial demography. Sci Data 2017;4:170004. https://doi.org/10.1038/sdata.2017.4.
[54] Tegou L-I, Polatidis H, Haralambopoulos DA. Environmental management framework for wind farm siting: Methodology and case study. J Environ Manage 2010;91(11):2134–47. https://doi.org/10.1016/j.jenvman.2010.05.010.
[55] Höfer T, Sunak Y, Siddique H, Madlener R. Wind farm siting using a spatial Analytic Hierarchy Process approach: A case study of the Städteregion Aachen. Appl Energy 2016;163:222–43. https://doi.org/10.1016/j.apenergy.2015.10.138.
[56] Uyan M. GIS-based solar farms site selection using analytic hierarchy process (AHP) in Karapinar region. Konya/Turkey. Renewable and Sustainable Energy Reviews 2013;28:11–7. https://doi.org/10.1016/j.rser.2013.07.042.
[57] Amankwah E. Environmental Impact Assessment (EIA); A Useful Tool to Address Climate Change in Ghana. IJEPP 2013;1:94. https://doi.org/10.11648/j.ijepp.20130104.18.
[58] MacQueen J. Some methods for classification and analysis of multivariate observations. University of California Press 1967;vol. 1.
[59] Kaufman L, Rousseeuw PJ. Partitioning around medoids (program pam). Finding Groups in Data: An Introduction to Cluster Analysis 1990;344:68–125.
[60] Colak HE, Memisoglu T, Gercek Y. Optimal site selection for solar photovoltaic (PV) power plants using GIS and AHP: A case study of Malatya Province. Turkey. Renewable Energy 2020;149:565–76. https://doi.org/10.1016/j.renene.2019.12.078.
[61] Ozdemir S, Sahin G. Multi-criteria decision-making in the location selection for a solar PV power plant using AHP. Measurement 2018;129:218–26. https://doi.org/10.1016/j.measurement.2018.07.020.
[62] S.M. Habib A. El-Raie Emam Suliman A.H. Al Nahry E.N. Abd El Rahman 18 2020 100313 10.1016/j.rsase.2020.100313.
[63] Goepel KD. New AHP Excel template with multiple inputs – BPMSG n.d. https://bpmsg.com/new-ahp-excel-template-with-multiple-inputs/ (accessed July 19, 2021).
[64] Goepel KD. Implementation of an Online Software Tool for the Analytic Hierarchy Process (AHP-OS). International Journal of the Analytic Hierarchy Process 2018; 10. https://doi.org/10.13033/ijahp.v10i3.590.
[65] Goepel KD. Implementing the Analytic Hierarchy Process as a Standard Method for Multi-Criteria Decision Making in Corporate Enterprises – a New AHP Excel Template with Multiple Inputs, 2013. https://doi.org/10.13033/isahp.y2013.047.
[66] Al Garni HZ, Awasthi A. Solar PV power plant site selection using a GIS-AHP based approach with application in Saudi Arabia. Appl Energy 2017;206:1225–40. https://doi.org/10.1016/j.apenergy.2017.10.024.
[67] Agyekum EB, Nutakor C. Feasibility study and economic analysis of stand-alone hybrid energy system for southern Ghana. Sustainable Energy Technol Assess 2020;39:100695. https://doi.org/10.1016/j.seta.2020.100695.
[68] Suzer AE, Atasoy VE, Ekici S. Developing a holistic simulation approach for parametric techno-economic analysis of wind energy. Energy Policy 2021;149: 112105. https://doi.org/10.1016/j.enpol.2020.112105.
[69] Bilal B, Adjallah KH, Yetilmezsoy K, Bahramian M, Kıyan E. Determination of wind potential characteristics and techno-economic feasibility analysis of wind turbines for Northwest Africa. Energy 2021;218:119558. https://doi.org/10.1016/j.energy.2020.119558.
[70] Mahian O, Javidmehr M, Kasaeian A, Mohasseb S, Panahi M. Optimal sizing and performance assessment of a hybrid combined heat and power system with energy storage for residential buildings. Energy Convers Manage 2020;211:112751. https://doi.org/10.1016/j.enconman.2020.112751.

[71] Ali F, Ahmar M, Jiang Y, AlAhmad M. A techno-economic assessment of hybrid energy systems in rural Pakistan. Energy 2021;215:119103. https://doi.org/10.1016/j.energy.2020.119103.
[72] Agyekum EB, Adebayo TS, Bekun FV, Kumar NM, Panjwani MK. Effect of Two Different Heat Transfer Fluids on the Performance of Solar Tower CSP by Comparing Recompression Supercritical CO2 and Rankine Power Cycles. China. Energies 2021;14:3426. https://doi.org/10.3390/en14123426.
[73] Ruegg RT, Marshall HE. In: Building Economics: Theory and Practice. Boston, MA: Springer US; 1990. p. 92–104. https://doi.org/10.1007/978-1-4757-4688-4_7.
[74] Agyekum EB, Afornu BK, Ansah MNS. Effect of Solar Tracking on the Economic Viability of a Large-Scale PV Power Plant. Environmental and Climate Technologies 2020;24:55–65. https://doi.org/10.2478/rtuect-2020-0085.
[75] Agyekum EB. Techno-economic comparative analysis of solar photovoltaic power systems with and without storage systems in three different climatic regions. Ghana. Sustainable Energy Technologies and Assessments 2021;43:100906. https://doi.org/10.1016/j.seta.2020.100906.
[76] Azimoh CL, Klintenberg P, Wallin F, Karlsson B, Mbohwa C. Electricity for development: Mini-grid solution for rural electrification in South Africa. Energy Convers Manage 2016;110:268–77. https://doi.org/10.1016/j.enconman.2015.12.015.
[77] Energy Commission. 2020 Energy (Supply and Demand) Outlook for Ghana 2020.
[78] Kousky C, Ritchie L, Tierney K, Lingle B. Return on investment analysis and its applicability to community disaster preparedness activities: Calculating costs and returns. Int J Disaster Risk Reduct 2019;41:101296. https://doi.org/10.1016/j.ijdrr.2019.101296.
[79] Zamfir M, Manea MD, Ionescu L. Return on Investment – Indicator for Measuring the Profitability of Invested Capital. Valahian Journal of Economic Studies 2016;7:79–86. https://doi.org/10.1515/vjes-2016-0010.
[80] Said Z, Mehmood A, Waqas A, Amine Hachicha A, Loni R. Central versus off-grid photovoltaic system, the optimum option for the domestic sector based on techno-economic-environmental assessment for United Arab Emirates. Sustainable Energy Technol Assess 2021;43:100944. https://doi.org/10.1016/j.seta.2020.100944.
[81] Global Wind Atlas n.d. https://globalwindatlas.info/ (accessed November 17, 2020).
[82] Maatallah T, Ghodhbane N, Ben NS. Assessment viability for hybrid energy system (PV/wind/diesel) with storage in the northernmost city in Africa, Bizerte. Tunisia. Renewable and Sustainable Energy Reviews 2016;59:1639–52. https://doi.org/10.1016/j.rser.2016.01.076.
[83] Arefin S, Das N. Optimized Hybrid Wind-Diesel Energy System with Feasibility Analysis. Technol Econ Smart Grids Sustain Energy 2017;2:9. https://doi.org/10.1007/s40866-017-0025-6.
[84] Shezan SKA, Lai CY. Optimization of hybrid wind-diesel-battery energy system for remote areas of Malaysia. Australasian Universities Power Engineering Conference (AUPEC) 2017;2017:1–6. https://doi.org/10.1109/AUPEC.2017.8282390.
[85] Li C, Zhou D, Wang H, Lu Y, Li D. Techno-economic performance study of stand-alone wind/diesel/battery hybrid system with different battery technologies in the cold region of China. Energy 2020;192:116702. https://doi.org/10.1016/j.energy.2019.116702.
[86] Rad MAV, Ghasempour R, Rahdan P, Mousavi S, Arastounia M. Techno-economic analysis of a hybrid power system based on the cost-effective hydrogen production method for rural electrification, a case study in Iran. Energy 2020;190:116421. https://doi.org/10.1016/j.energy.2019.116421.
[87] Azerefegn TM, Bhandari R, Ramayya AV. Techno-economic analysis of grid-integrated PV/wind systems for electricity reliability enhancement in Ethiopian industrial park. Sustainable Cities and Society 2020;53:101915. https://doi.org/10.1016/j.scs.2019.101915.
[88] Das BK, Hoque N, Mandal S, Pal TK, Raihan MA. A techno-economic feasibility of a stand-alone hybrid power generation for remote area application in Bangladesh. Energy 2017;134:775–88. https://doi.org/10.1016/j.energy.2017.06.024.
[89] Kotb KM, Elkadeem MR, Elmorshedy MF, Dán A. Coordinated power management and optimized techno-enviro-economic design of an autonomous hybrid renewable microgrid: A case study in Egypt. Energy Convers Manage 2020;221:113185. https://doi.org/10.1016/j.enconman.2020.113185.
[90] Salisu S, Mustafa MW, Olatomiwa L, Mohammed OO. Assessment of technical and economic feasibility for a hybrid PV-wind-diesel-battery energy system in a remote community of north central Nigeria. Alexandria Engineering Journal 2019;58(4):1103–18. https://doi.org/10.1016/j.aej.2019.09.013.
[91] Esan AB, Agbetuyi AF, Oghorada O, Ogbeide K, Awelewa AA, Afolabi AE. Reliability assessments of an islanded hybrid PV-diesel-battery system for a typical rural community in Nigeria. Heliyon 2019;5(5):e01632. https://doi.org/10.1016/j.heliyon.2019.e01632.